Design Architecture of ZFNs, TALENs and CRISPR-Cas9: An In-Silico Case Study at One BCL11A Coding Exon, with a 45-Exon Site-Count Extension

A consolidated comparative manuscript combining the CRISPR-Cas9, TALEN and ZFN design walkthroughs, the candidate-site script and its multi-locus run

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Tools accessed 24 October 2026; scripts run 25 October 2026 and later; genome build GRCh38/hg38. All numerical values are computational predictions or rule-based counts; no experiment was performed.

Abstract

Background. Zinc finger nucleases (ZFNs) and transcription activator-like effector nucleases (TALENs) read DNA through engineered protein arrays, whereas CRISPR-Cas9 is aimed by a short guide RNA that pairs with DNA by Watson-Crick rules. This paper documents what each platform requires at one concrete target and what free in-silico tools and a transparent script can say about it.

Approach. We designed all three nuclease types against one coding exon of human *BCL11A* (GRCh38, amplicon chr2:60,462,161-60,462,397, 237 bp) using CHOPCHOP, CRISPOR v5.2 and the UCSC Genome Browser. CRISPR-Cas9 and TALEN designs came from CHOPCHOP. No ZFN design tool could be reached, so the ZFN pair comes from a rule-based scan. A short Python script (S1) applied explicit rules to count candidate sites for each platform. To test whether the single-window counts generalise, the same script was run on 45 randomly sampled coding-exon windows of 237 bp (RefSeq NM_ transcripts, hg38, seed 1).

Findings. The selected SpCas9 guide (spacer 5'-CCTGAAGGGATACCAACCCG-3', PAM CGG) had a CHOPCHOP score of 76.99 and a CRISPOR Doench '16 score of 77 (probably the same model), no predicted look-alike sites at 0-2 mismatches, and a Cas9-specific repair model predicting 83.91% frameshift outcomes. A CHOPCHOP TALEN pair in the same exon spans 51 bp with two 18-RVD arrays (36 RVDs as listed). A rule-based ZFN pair exists under a strict GNNx3 rule. In the BCL11A window the script counted 32 Cas9 guides, 47 or 58 TALEN candidates (two rules) and 1 or 19 ZFN pairs (two rules). Across 45 exon windows the median densities were 143.5 (SpCas9), 126.6 and 189.9 (TALEN, tool-like and broad rules) and 97.0 (relaxed ZFN) candidates per kb, while the median strict-ZFN count was 0.0 per kb. The BCL11A window was therefore richer in TALEN sites than a typical window and richer in strict ZFN sites than the median.

Interpretation. The number of units that must be specified differs by architecture: one ~20-nt spacer for Cas9, about 34-36 RVDs in two proteins for TALEN and about six fingers in two proteins for ZFN. This follows from how each platform works and does not depend on the locus. Candidate-site density was of the same order for Cas9 and TALEN across loci, so the data do not show that Cas9 has more target sites; strict modular ZFN sites were rare. Claims about efficiency, specificity, cost and multiplexing rest on the literature and were not tested.

Wet-lab validation is required before any such conclusion.

Keywords: CRISPR-Cas9; ZFN; TALEN; BCL11A; in-silico case study; guide RNA design; repeat variable di-residue (RVD); candidate-site count; genome editing

1. Introduction and Theoretical Framework

1.1 Why targeted editing needs a programmable nuclease

Classical gene therapy adds a functional gene copy, usually through an integrating viral vector, at an uncontrolled position. Integration beside proto-oncogenes caused leukaemia in X-linked severe combined immunodeficiency (X-SCID) gene-therapy trials [39]. Targeted genome editing instead changes the endogenous locus itself: it disrupts a gene, corrects a mutation in place or inserts a transgene at a chosen position, so the gene stays under its own regulatory control. The obstacle to early targeted modification was the rarity of homologous recombination (HR); a site-specific double-strand break (DSB) raises HR frequency by orders of magnitude [36,37]. The break is the catalyst, because it recruits the cell's repair pathways to the chosen site. The problem of genome editing therefore reduces to an engineering question: how readily can a nuclease be aimed at an arbitrary sequence? This paper examines one facet of that question, the architecture of the design problem, at one target.

1.2 Three generations of programmable nucleases

Early targeted recombination relied on gene targeting in mouse embryonic stem cells and on natural homing endonucleases (meganucleases) such as I-SceI. Their long recognition sequences make them rare cutters, but DNA binding and catalysis overlap in one protein, so each new target needed extensive protein engineering [38]. The solution was a chimeric nuclease that separates recognition from catalysis [32,33].

ZFNs. Kim, Cha and Chandrasegaran fused zinc finger DNA-binding modules to the cleavage domain of the Type IIS enzyme FokI [1]. Each Cys2His2 zinc finger recognises about 3 bp, so a 3-finger array reads a 9-bp half-site. FokI is active only as a dimer, so two monomers must bind opposite strands separated by a short spacer (5-7 bp was used here). **TALENs.** TAL effectors from plant-pathogenic *Xanthomonas* carry tandem repeats in which two variable residues, the repeat variable di-residue (RVD), specify one base. The 1:1 code was decoded in 2009 [5,6], and TALE-FokI fusions followed in 2010-2011 [7,8,9]. **CRISPR-Cas9.** In 2012 Jinek and colleagues and Gasiunas and colleagues showed that Cas9 can be programmed in vitro by a crRNA:tracrRNA duplex, and that a single guide RNA (sgRNA) preserves activity [11,12]; in 2013 Cong and Mali extended this to human cells [13,14] (see also [34]).

Table 1. Evolutionary trajectory of programmable nucleases (from the cited primary literature and reviews).

Year	Platform	Key milestone	Recognition principle
1996	ZFN	Zinc finger-FokI hybrid nuclease [1]	Protein-DNA; ~3 bp per finger
2009	TAL code	1 repeat : 1 base cipher decoded [5,6]	Protein-DNA; RVD at residues 12-13
2010-2011	TALEN	TALE-FokI nucleases; Golden Gate assembly (~5 days) [7,8,9]	Protein-DNA; 1 bp per repeat
2012	CRISPR-Cas9	Programmable dual-RNA then single-guide Cas9 [11,12]	RNA-DNA base pairing; 20 nt + PAM
2013	CRISPR-Cas9	Genome editing in human cells [13,14]	RNA-DNA base pairing; 20 nt + PAM

Year	Platform	Key milestone	Recognition principle
2021	Clinical Cas9	Cas9 editing of a BCL11A enhancer in sickle cell disease and beta-thalassaemia [22] (exa-cel, Casgevy)	Cas9 targeting a regulatory element, not a coding exon

1.3 Mechanics of recognition: protein-DNA versus RNA-DNA

Zinc fingers. A ZFN monomer is a tandem array of fingers, each contacting a three-base subsite. Fingers are not independent: neighbouring fingers influence each other, subsites can overlap, and a finger that works alone may bind poorly beside others. This context dependence is why modular assembly can fail and why selection-based methods were developed [29,30]. For rule-based design we used the canonical GNN rule (each triplet begins with G), the subset for which pre-characterised modules exist [2,3].

TALE repeats. The TALE DNA-binding domain consists of tandem repeats of 33-35 amino acids that are nearly identical except at positions 12 and 13, the RVD. Residue 13 is the base-specifying residue and contacts the base in the major groove; residue 12 stabilises the presenting loop [26]. The repeats stack into a right-handed superhelix around the major groove, and target sites are preceded by a 5' thymine (T0) read by cryptic N-terminal repeats. CHOPCHOP's mapping is A = NI, C = HD, T = NG, G = NH (NN reads G and also A, and is therefore less specific).

Table 2. Common RVDs and base preferences. All RVD arrays here follow the CHOPCHOP convention (A = NI, C = HD, T = NG, G = NH).

RVD	Preferred base	Structural basis (from the cited reviews)
NI	A	Asn13 hydrogen-bonds with adenine
HD	C	Asp13 hydrogen-bonds with the cytosine amino group
NG	T	Gly13 makes van der Waals contact with the thymine methyl group
NN	G / A	Asn13 contacts guanine and also adenine
NH	G	Used by CHOPCHOP for guanine; more G-specific than NN
NS	any	Degenerate; recognises any base

Guide RNA. In the native Type II system the crRNA contributes a ~20-nt spacer-derived guide and the tracrRNA base-pairs with the crRNA repeat to form a scaffold recognised by Cas9. The engineered sgRNA has a 5' spacer of 20 nt complementary to the target strand and a 3' scaffold that binds Cas9; only the spacer changes to retarget the nuclease. Cas9 first scans DNA for a protospacer adjacent motif (PAM; 5'-NGG-3' for *Streptococcus pyogenes* Cas9, SpCas9). Local melting follows, and the 8-12 PAM-proximal nucleotides (the seed) pair first. Base pairing then proceeds toward the distal end to form an R-loop; the HNH and RuvC-like domains cut the two strands, producing a blunt DSB about 3 bp upstream of the PAM [15].

The practical consequence is a difference in how sequence information is handled. For Watson-Crick pairing, the sequence-to-specificity rule is the same for every target and is captured by simple scoring models, whereas protein-DNA recognition is context-dependent (ZFNs) or needs long, near-identical repeat arrays (TALENs). A ZFN pair reads 18 bp (2 x 9), a TALEN pair about 30-40 bp of arm sequence plus the 5' T, and SpCas9 reads 20 nt plus a 3-bp PAM from one RNA.

1.4 DSB repair: NHEJ, MMEJ and HDR

All three platforms hand the break to the same cellular machinery, so the editing outcome is largely platform-independent. Non-homologous end joining (NHEJ) is the default pathway, active throughout the cell cycle; it ligates processed ends without a template, often leaving small insertions or deletions (indels). In a coding exon, indels that are not multiples of three cause frameshifts. Microhomology-mediated end joining (MMEJ) is an alternative error-prone route that anneals short flanking homologies and yields defined deletions. Homology-directed repair (HDR) is template-dependent and high-fidelity but restricted largely to S and G2 phases.

Table 3. Repair pathways engaged by nuclease-induced DSBs.

Pathway	Cell-cycle activity	Template	Typical outcome	Main use
NHEJ	All phases	None	Indels, frameshifts	Gene knockout
MMEJ	Mostly S/G2	Short microhomologies	Defined deletions	Deletions
HDR	S / G2	Sister chromatid or donor DNA	Precise substitution or insertion	Correction, knock-in

1.5 Thesis, scope and objectives

Working thesis. ZFNs and TALENs supplied the architecture on which gene targeting was built: a programmable DNA-binding module joined to a nuclease. CRISPR-Cas9 relocates recognition from a newly engineered protein to a short guide RNA, so its target is specified by a ~20-nt spacer beside a PAM. We argue that this RNA-guided mechanism makes the design problem architecturally simpler than for protein-guided nucleases. Whether it also outperforms them in efficiency, specificity, cost and multiplexing cannot be decided from in-silico data; for those axes the paper relies on published literature and says so.

Objectives. (1) Design the three nuclease types against the same exon and record what each tool or rule produces. (2) Report predicted on-target quality and off-target output where a tool provides it. (3) Count candidate sites for all three platforms under explicit rules, first in the benchmark window and then in 45 other coding-exon windows. (4) State what the data can and cannot support.

Note on the locus. *BCL11A* was chosen because it is well known in genome editing: the approved therapy exagamglogene autotemcel (Casgevy) edits an erythroid enhancer of *BCL11A*, not the coding exon [21,22,24]. Our designs target a coding exon, so they are illustrative knock-out designs, not therapeutic guides. The exon also offered candidate sites for all three platforms within one 237-bp PCR amplicon, which allowed a like-for-like comparison.

2. Materials and Computational Methodology

2.1 Study design and limits

The target region is held constant and the platform varies. This removes local sequence context as a confounder for the site-count analysis only. It cannot separate platform properties from properties of this one exon. Everything was done in a web browser or with short scripts on a laptop (Windows, VS Code, Python); nothing was built or tested in the laboratory, and every numerical output is labelled predicted or rule-based.

2.2 Target sequence

The gene is *BCL11A* (human chromosome 2p16.1), GRCh38/hg38. The region chr2:60,462,161-60,462,397 is the CHOPCHOP-suggested PCR product (Pair 1). The sequence was retrieved from the UCSC sequence service (start 60462160, end 60462397, 0-based half-open), returning a 237-nt string that starts CCTCTCTCGATACTGATCCTGG and ends GTTGCAAGTTGTAC, in the same orientation as the CHOPCHOP output (Figures 16-18). It was verified to contain the CRISPR spacer and the TALEN pair. The one-base difference from the 236-bp product in the primer table is a counting convention. Whether the whole exon is coding, rather than partly UTR, was not confirmed on the UCSC CDS track.

2.3 CRISPR-Cas9 design and evaluation

Guides were ranked in CHOPCHOP (hg38, SpCas9, knock-out) [16,54]. Rank 1 was opened for the gene map, predicted cut, suggested primers and the Shen et al. (2018) repair-outcome model [18]. Pair 1 was chosen as the validation amplicon (equal Tm 60.2 C, no primer off-targets, 236-bp product suited to a T7E1 assay). The 120-bp exon sequence (chr2:60,462,222-60,462,341) was scored in CRISPOR v5.2 (hg38, 20bp-NGG) [17], which reports MIT and CFD specificity, Doench '16, Moreno-Mateos and Rule Set 3 efficiency scores, out-of-frame and Lindel scores, and off-target counts at 0-4 mismatches. We did not record which efficiency model CHOPCHOP used. Its Cas9 knock-out default is, to our understanding, the Doench 2016 model [19], which is also CRISPOR's 'Doench '16' score, so agreement between 76.99 and 77 is a consistency check, not independent validation. The Moreno-Mateos (64) and Rule Set 3 (81) scores differ from it by up to 17 points, so scores should not be read as exact. Predicted efficiency is an on-target activity score, not a percentage of edited cells.

2.4 TALEN design

CHOPCHOP in TALEN mode (hg38, knock-out) ranks target pairs [54]. The pair was chosen as the highest-ranked candidate in the same exon as the guide with 0 predicted off-target pairs (Rank 3, chr2:60,462,267). Because 0 off-target pairs was a selection criterion, it is not reported as a finding. Rank 1 (chr2:60,553,212) lies in a different exon and was not used.

Every CHOPCHOP target string shown begins with T and ends with A. We read this as T0 + left arm + spacer + right arm + terminal A, the convention of TALE-NT [55]: T0 at amplicon position 107, arms of 17 nt, a 15-bp spacer and a terminal A at position 157. Treating the first T as the first repeat instead would need a T at position 106 and an A at position 158; the bases there are C and G. The T0 reading and the orientation of the right-hand array remain to be confirmed in the tool documentation or with TALE-NT.

2.5 ZFN design (rule-based, no tool output)

No ZFN design tool could be used. The one design-tool link tried returned a 403 Forbidden page from an unrelated university site, so ZiFiT [3] and ZFNGenome [4] were not reached, and no failure of those services is claimed. The pair was found by scanning the 237-nt sequence: every 9-bp window on the top strand was tested as a left half-site, and a right half-site on the opposite strand was tested after a 5-, 6- or 7-bp spacer. A half-site was accepted if it had the form GNN-GNN-GNN, the most restrictive rule, chosen because pre-characterised modules exist for it [2,3]. GNN-only design excludes finger types that real design uses (ANN, CNN, TNN modules, longer arrays, selection methods) [29,30]. No tool score or off-target prediction was obtained, so ZFN specificity is not assessed.

2.6 Candidate-site count (script S1)

The script site_count.py (Supplement S1; Figures 21 and 22) applies explicit rules to a DNA sequence. **SpCas9:** a 20-nt protospacer followed by NGG on either strand, wholly inside the window. **TALEN 'tool-like':** T0 + 17-nt arm + 14-16-bp spacer + 17-nt arm + terminal A, counted as distinct pairs (spacer variants counted separately). **TALEN 'broad':** arms of 15-20 nt and spacer of 12-30 bp, counted as distinct T0 positions with at least one valid partner. **ZFN:** left half-site on the top strand, right half-site on the bottom strand, spacer 5-7 bp; 'strict' = GNNx3, 'relaxed' = at least 2 of 3 triplets GNN, plus a 4-finger GNNx4 check. These are rule-based counts, not tool outputs, and apply none of the tools' filters (RVD composition, uniqueness, off-target screening). Only the Cas9 rule was checked against a tool: the script returned 18 guides in the 120-bp CRISPOR window, matching CRISPOR's 18. For TALEN the script recovered the CHOPCHOP Rank 3 pair, which tests recall but not precision, and for ZFN it recovered the single strict pair. The script was run on the author's laptop and reproduced every single-window count (Figure 22); an independent re-implementation of the same rules gave identical counts.

2.7 Multi-locus extension (45 coding-exon windows)

To test whether the single-window counts generalise, the same script was run on windows from other loci (Supplements S2 and S3). RefSeq transcripts were downloaded from the UCSC Table Browser (hg38, NCBI RefSeq, BED format; 197,654 transcripts). make_exons.py kept curated protein-coding transcripts (accession NM_) on chromosomes 1-22, X and Y (69,706 transcripts), clipped each exon to its coding part (thickStart to thickEnd) and counted exons with identical coordinates once, giving 216,361 unique coding exons, of which 23,754 were at least 237 bp. Five hundred exons were drawn with Python's random.seed(1) and random.sample [Python version: confirm with python --version]. Exons shorter than 237 bp were skipped, and one 237-bp window was taken from each of the remaining exons: fetch_windows.py cuts the central 237 bp of the exon (window start = midpoint - 118) and downloads it from the UCSC sequence service. The 45 windows retained were counted with site_count.py, and the median candidates per kb across windows is reported for each rule. Downloads from the laptop initially failed with a certificate error (CERTIFICATE_VERIFY_FAILED) and were completed after supplying Python with a trusted certificate store.

2.8 Reporting conventions and use of AI tools

The pipeline is deterministic, so no inferential statistics were applied; counts and scores are reported as given by each tool or script, and values that two tools report differently are shown side by side. Derived numbers are labelled derived. Figures 1, 2, 3, 19 and 20 are schematics or charts generated from the reported coordinates and values; all other figures are unmodified screenshots or, where stated, a photograph of the screen. **AI assistance statement [adapt to the target journal's policy]:** an AI assistant (Claude, Anthropic) helped organise the workflow, derive RVD strings, write the scripts and draft the text. The authors are responsible for all content, and every rule-based count can be reproduced with the supplementary scripts.

Table 4. Software and settings. Web resources accessed 24 October 2026; scripts run 25 October 2026 and later.

Resource	Version / source	Settings	Role
CHOPCHOP (CRISPR mode)	Web server, v3 [16]	hg38; SpCas9; knock-out; BCL11A	Guide ranking, primers, off-targets, Shen et al. repair model

Resource	Version / source	Settings	Role
CRISPOR	v5.2 [17]	hg38; 20bp-NGG; 120 bp input	Second scoring tool (MIT, CFD, Doench '16, Lindel)
CHOPCHOP (TALEN mode)	Web server [54]	hg38; TALEN; knock-out	Pair ranking, RVD arrays, pair off-target check
UCSC Genome Browser and sequence service	GRCh38/hg38	chr2:60,462,161-60,462,397	Coordinates and 237-nt sequence
UCSC Table Browser	hg38, NCBI RefSeq	BED output, 197,654 transcripts	Exon source for the multi-locus run
site_count.py (S1)	This work	Rules in Section 2.6	Rule-based site counts and ZFN scan
fetch_windows.py (S2)	This work	Central 237-bp window per exon	Sequence download
make_exons.py (S3)	This work	NM_ only; chr1-22, X, Y; CDS part; unique exons	Exon list for sampling

3. Results

All three designs lie within the 237-bp window and within about 30 bp of each other (Figure 1). Every value below is a predicted or rule-based result from the named tool or script. The designs are at different positions in one exon, not at one identical site. Screenshots of tool output are collected in Section 6 (Figures 4-18).

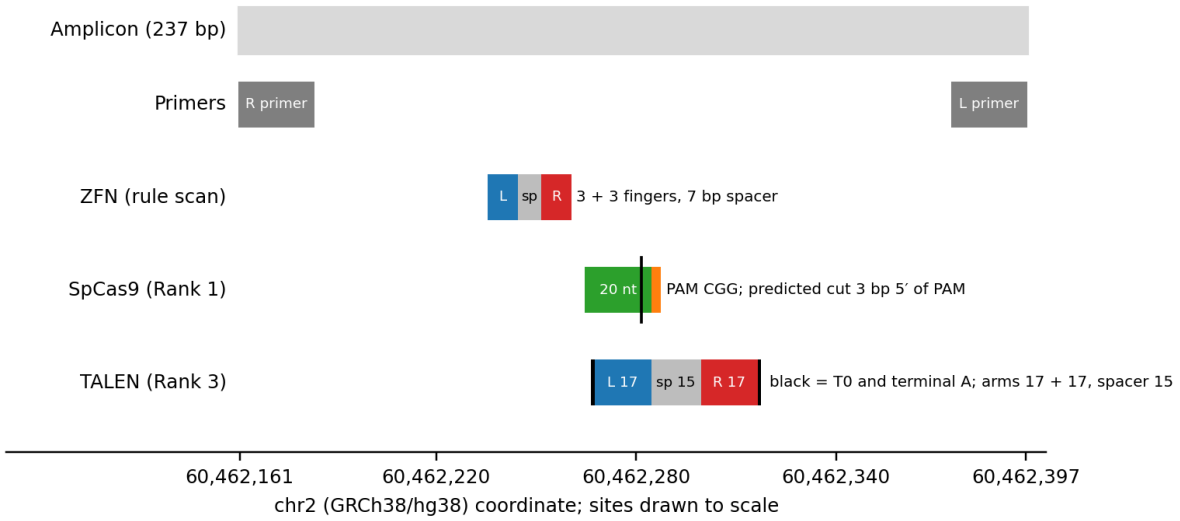


Figure 1. Co-location of the three designs on the amplicon, drawn to scale from the reported coordinates (chr2:60,462,161-60,462,397, GRCh38/hg38). ZFN: candidate pair from a rule-based scan. CRISPR-Cas9: Rank 1 SpCas9 spacer (chr2:60,462,265-84) with PAM CGG; the cut (between 60,462,281 and 60,462,282) is derived from the 3-bp-upstream-of-PAM rule. TALEN: CHOPCHOP Rank 3 target string read as T0 (60,462,267) + 17-nt left arm + 15-bp spacer + 17-nt right arm + terminal A (60,462,317).

3.1 CRISPR-Cas9

CHOPCHOP drew the gene map with five RefSeq models and listed ranked guides (Figure 4). Rank 1 combined the highest predicted efficiency with ideal GC content, no self-complementarity and no predicted look-alike sites at 0-2 mismatches. The Rank 1 spacer is 5'-CCTGAAGGGATACCAACCCG-3' with PAM CGG (forward strand, chr2:60,462,265); in

amplicon coordinates the spacer occupies positions 105-124 and the PAM 125-127. The predicted cut lies between chr2:60,462,281 and 60,462,282 (derived). The guide sits in a coding exon shared by most BCL11A transcript models (Figure 5).

Table 5. Top five CHOPCHOP guides for BCL11A (SpCas9, knock-out, hg38; accessed 24 October 2026). Efficiency is a predicted on-target score.

Rank	Target (PAM)	Location (hg38)	Strand	GC %	Self-compl.	MM0/1/2/3	Predicted efficiency
1	CCTGAAGGGATACCAACCCG (CGG)	chr2:60462265	+	60	0	0/0/0/1	76.99
2	GATAAACAATCGTCATCCTC (TGG)	chr2:60546015	+	40	0	0/0/0/1	55.03
3	GATACCAACCCGCGGGGTCA (GGG)	chr2:60462273	+	65	1	0/0/0/1	53.70
4	TGAAGGGATACCAACCCGCG (GGG)	chr2:60462267	+	60	0	0/0/0/2	71.78
5	TGAACCAGACCACGGCCCGT (TGG)	chr2:60546247	-	65	1	0/0/0/2	56.03

Off-target output and cross-tool agreement. CHOPCHOP lists one look-alike at 3 mismatches (chr5:82,019,991; Figure 6). CRISPOR places the same guide at the identical coordinate and reports MIT specificity 94, CFD specificity 98 and off-target counts of 0, 0, 0, 4 and 44 at 0-4 mismatches (48 in total; Figures 8, 9). The tools agree on no look-alikes at 0-2 mismatches but report 1 versus 4 sites at 3 mismatches; the reason was not checked (possible causes include different genome indexes or mismatch-counting rules), so both numbers are reported.

Table 6. Side-by-side predictions for the Rank 1 guide (hg38; accessed 24 October 2026).

Check	CHOPCHOP	CRISPOR v5.2	Comment
On-target score	76.99	Doench '16 = 77; Moreno-Mateos 64; Rule Set 3 = 81	Likely the same model; consistency, not independence
Off-targets, 0-2 mismatches	None	None (0/0/0)	Agree
Off-targets, 3 mismatches	1 (chr5:82,019,991)	4	Differ; reason not checked
Off-targets, 4 mismatches	not listed	44 (48 in total with 3 mm)	CRISPOR only
Specificity scores	-	MIT 94; CFD 98	CRISPOR only
Frameshift	83.91% (Shen et al.)	Out-of-frame 73; Lindel 84	Different models, same direction

A neighbouring guide, TGAAGGGATACCAACCCGCG (PAM GGG), has better CRISPOR specificity (MIT 96; 25 off-targets in total) but a lower Doench '16 score (72); a third neighbour, CTGAAGGGATACCAACCCGC, scored MIT 94, CFD 96, Doench '16 = 57. A second guide near this site would be a sensible backup in a real experiment.

Predicted repair outcomes. The Shen et al. model embedded in CHOPCHOP predicts 83.91% frameshift outcomes for Rank 1 (+1 frame 47.58%, +2 frame 36.33%, in-frame 16.09%), with 77.42% microhomology deletions, 16.12% microhomology-less deletions and 6.46% 1-bp insertions, and a precision score of 0.47 (Figures 2 and 7; Table 7). These are shares of model-predicted outcomes, not fractions of cells knocked out. The model was trained on Cas9 blunt-end breaks and does not apply to FokI-generated breaks, so it is a Cas9 output only. 'Microhomology deletion' is a sequence class in the model, not a measured

repair pathway.

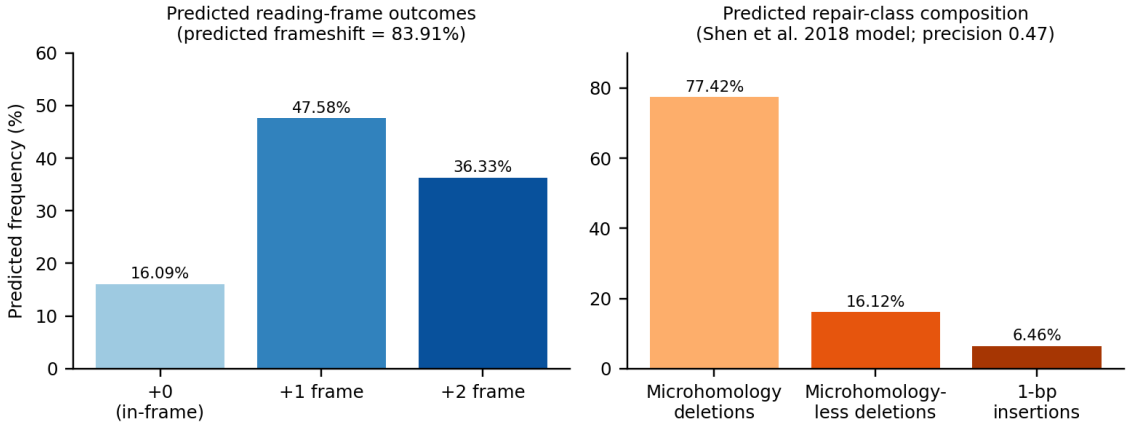


Figure 2. Predicted repair outcomes after Cas9 cleavage at the Rank 1 site (Shen et al. 2018 model, as shown by CHOPCHOP). Left: reading-frame distribution. Right: repair-class composition. Model predictions, not experimental measurements.

Table 7. Predicted repair statistics for the Rank 1 guide (CHOPCHOP; accessed 24 October 2026). The three frame classes sum to 100.00% (16.09 + 47.58 + 36.33) and the three repair classes sum to 100.00% (77.42 + 16.12 + 6.46).

Statistic (Shen et al. 2018 model)	Predicted value
Frameshift frequency	83.91%
Frame +0 (in-frame) / +1 / +2	16.09% / 47.58% / 36.33%
1-bp insertion frequency	6.46%
Highest deletion / highest insertion frequency	19.10% / 3.95%
Microhomology deletion / microhomology-less deletion	77.42% / 16.12%
Precision score	0.47

Validation primers. Primer Pair 1 (left TACAACTTGCAAACAGCCATTC, chr2:60462375-60462397; right CCTCTCTCGATACTGATCCTGG, chr2:60462161-60462183; Tm 60.2 C for both; 236 bp; 0 predicted off-targets) spans all three designs, so one amplicon could test them (Figures 6 and 12).

3.2 TALEN

CHOPCHOP in TALEN mode returned two clusters of pairs, near chr2:60,553,000 and chr2:60,462,000 (Figure 10). Rank 1 lies in a different exon and was not used. The selected pair is Rank 3 (chr2:60,462,267; cluster 2; no restriction sites reported), whose target string starts 2 bp from the CRISPR spacer start (Figure 11). Rank 4 starts at the same position with a longer right site, so Ranks 3 and 4 are variants of one site, not two independent sites.

Table 8. Rank 3 TALEN pair, read with the T0/terminal-A convention (Section 2.4). The footprint including the spacer is 51 bp; the two arms alone are 34 nt.

Element	Sequence (5'-3', top strand)	Length	Amplicon positions	hg38 coordinates
T0	T	1	107	60,462,267
Left arm	GAAGGGATACCAACCCG	17	108-124	60,462,268-284
Spacer	CGGGGTCAGGGGACT	15	125-139	60,462,285-299
Right arm (opposite strand)	TCCGTGTTCGCTTTCTA	17	140-156	60,462,300-316

Element	Sequence (5'-3', top strand)	Length	Amplicon positions	hg38 coordinates
Terminal A	A	1	157	60,462,317
Footprint	-	51	107-157	60,462,267-317

RVD arrays. CHOPCHOP lists 18 RVDs per array (36 in total). The first 12 per array are visible in the screenshot in Figure 13; the complete 18-per-side arrays were read from a photograph of the same table taken on 24 October 2026 (Supplementary Figure S2) and agree repeat for repeat with the arrays derived from the displayed string with A = NI, C = HD, T = NG, G = NH (Table 9). If the leading T of each array is the T0 read by the N-terminal domain rather than a counted repeat, 34 RVDs are needed instead of 36. The listed right array follows the top-strand string TCCGTG...; a TALE binding the bottom strand would need RVDs for the reverse complement. Read from the bottom strand, the right arm is T0 followed by TAGAAAGCGAACACGGA, which corresponds to NG NI NH NI NI NI NH HD NH NI NI HD NI HD NH NH NI. Which orientation CHOPCHOP uses must be confirmed before any array is ordered; this does not change the RVD count. The left array contains 6 NI, 5 NH, 5 HD and 2 NG; the right array 8 NG, 5 HD, 3 NH and 2 NI (counted from the listing).

Table 9. RVD arrays for the Rank 3 TALEN pair as listed by CHOPCHOP (A = NI, C = HD, T = NG, G = NH). Repeats 1-12 are visible in Figure 13; all 18 per side are visible in Supplementary Figure S2.

Repeats	Left TALEN (TGAAGGGATACCAACCCG)	Right TALEN (TCCGTGTTCGCTTTCTAA)
1-2	NG NH	NG HD
3-4	NI NI	HD NH
5-6	NH NH	NG NH
7-8	NH NI	NG NG
9-10	NG NI	HD NH
11-12	HD HD	HD NG
13-14	NI NI	NG NG
15-16	HD HD	HD NG
17-18	HD NH	NI NI

Off-target output. CHOPCHOP's pair box reads 'There are no off-targets' for this pair (Figure 12), which was a selection criterion and depends on the tool's mismatch limit. The half-site table lists 15/26 look-alikes at two mismatches (the meaning of '15/26' was not confirmed) and 0/0 at MM0 and MM1 (Figures 13-15). The tool's MM3 column reads 0/0, which we do not interpret as an absence of three-mismatch sites, because the tool may not search at that depth. TALEN half-site counts and Cas9 full-guide counts are different quantities and are not compared; a single look-alike cannot cause a cut because FokI needs both halves bound within the spacer range. TALENs can tolerate some mismatches [25], so '0 off-target pairs' is not proof of high fidelity. [Verify the mismatch-tolerance statement and reference 25 against the source.]

3.3 ZFN

No ZFN design tool output was obtained (Section 2.5). The 237-bp region was located in the UCSC Genome Browser (GRCh38/hg38, Figure 16), displayed at exactly 237 bp (Figure 17) and its sequence downloaded as JSON (Figure 18). Scanning for GNN-GNN-GNN half-site pairs separated by 5, 6 or 7 bp found three overlapping G-rich 9-mers on the top strand (amplicon positions 70, 73 and 76: GGAGAGGTG, GAGGTGGCT and GTGGCTGGG) and one

on the bottom strand (GGTGCAGAA, position 92); only one combination satisfied the spacer rule (Table 10, Figure 19). The scan was repeated by script and returned the same single pair.

Table 10. Candidate ZFN pair (rule-based scan; not a tool output). Each monomer is a 3-finger array reading 9 bp, fused to FokI.

	Left half-site	Spacer	Right half-site
Sequence (5'-3' on its own strand)	GTGGCTGGG	AAGGACA (7 bp)	GGTGCAGAA
Triplets (fingers)	GTG, GCT, GGG	-	GGT, GCA, GAA
Strand	Top	-	Bottom (read on the opposite strand)
Amplicon positions	76-84	85-91	92-100
Genomic position (hg38)	chr2:60462236-60462244	60462245-60462251	chr2:60462252-60462260
Finger modules	to be taken from the Zinc Finger Consortium archive or modular-assembly paper [2]	-	to be taken from the archive

The candidate is about 30 bp upstream of the Cas9 and TALEN designs, in the same exon: about 5 bp from the CRISPR guide start (60,462,265) and 31 bp from the TALEN T0 (60,462,267). Finger context effects make performance unpredictable from sequence alone [29,30]. One ZFN pair needs about six fingers (three per side) plus two FokI fusions, and every finger must still be tested for binding.

3.4 Candidate-site counts in the BCL11A window

Table 11. Rule-based counts in the 237-bp BCL11A window (script S1; reproduced on the author's laptop, Figure 22).

Platform / rule	Candidates in 237 bp	Counting unit
SpCas9, NGG, both strands	32	protospacer + PAM
TALEN, tool-like (17 + 17 arms, 14-16 bp spacer)	47	distinct pairs; spacer variants counted separately
TALEN, broad (15-20 arms, 12-30 bp spacer)	58	distinct T0 positions with at least one partner
ZFN, strict (GNNx3, 5-7 bp spacer)	1	pairs
ZFN, relaxed (at least 2 of 3 GNN)	19	pairs
ZFN, 4 fingers GNNx4	0	pairs

These counts are not directly comparable and are not used to rank platforms: the units differ (protospacers, pairs with duplicate spacer variants, anchors), the TALEN rules are looser than what a tool outputs, and no tool filters were applied. Only the Cas9 count was validated against a tool, so the TALEN counts are upper bounds. The literature estimate of roughly one TALEN site per ~35 bp [8] is far sparser than our TALEN count (about one pair per 5 bp), which confirms that our rules are permissive and not a measure of usable TALEN sites. The ZFN counts depend wholly on the rule: 1 under the strictest rule, 19 under a looser one, none with four strict fingers. GNN-only design is a property of our rule, not of ZFNs. The exon's G-richness raises NGG and GNN density directly, so these counts should not be assumed to hold at other loci, which is why the multi-locus run was done.

3.5 Multi-locus extension: 45 coding-exon windows

From 216,361 unique RefSeq coding exons (23,754 of at least 237 bp), 500 exons were sampled with seed 1, and 45 had a length of at least 237 bp and gave one central 237-bp window each (Section 2.7). Table 12 and Figure 20 give the median candidate density across the 45 windows next to the BCL11A window. SpCas9 density in BCL11A (135.0 per kb) was close to the median (143.5 per kb). TALEN density depended on the rule: the median was 126.6 per kb under the tool-like rule, below the Cas9 median, and 189.9 per kb under the broad rule, above it; BCL11A was richer in TALEN sites (198.3 and 244.7 per kb) than the typical window. The median strict ZFN count was 0.0 per kb, so the typical window held no GNNx3 pair, and the single BCL11A pair (4.2 per kb) was above the median. Under the relaxed ZFN rule the median density was 97.0 per kb, and no four-finger GNNx4 pair was found in the median window. The BCL11A window therefore was not typical for TALEN sites, and the Cas9 density was typical.

Table 12. Rule-based candidate density: BCL11A window versus the median of 45 coding-exon windows (hg38; script S1; seed 1). Medians are as reported from the run [confirm against results.csv; add the interquartile range and the share of windows with zero strict ZFN pairs].

Rule	BCL11A window, per kb (counts in Table 11)	Median of 45 exon windows, per kb
SpCas9, NGG	135.0	143.5
TALEN, tool-like	198.3	126.6
TALEN, broad	244.7	189.9
ZFN, strict GNNx3	4.2	0.0
ZFN, relaxed (at least 2 of 3 GNN)	80.2	97.0
ZFN, 4 fingers GNNx4	0.0	0.0

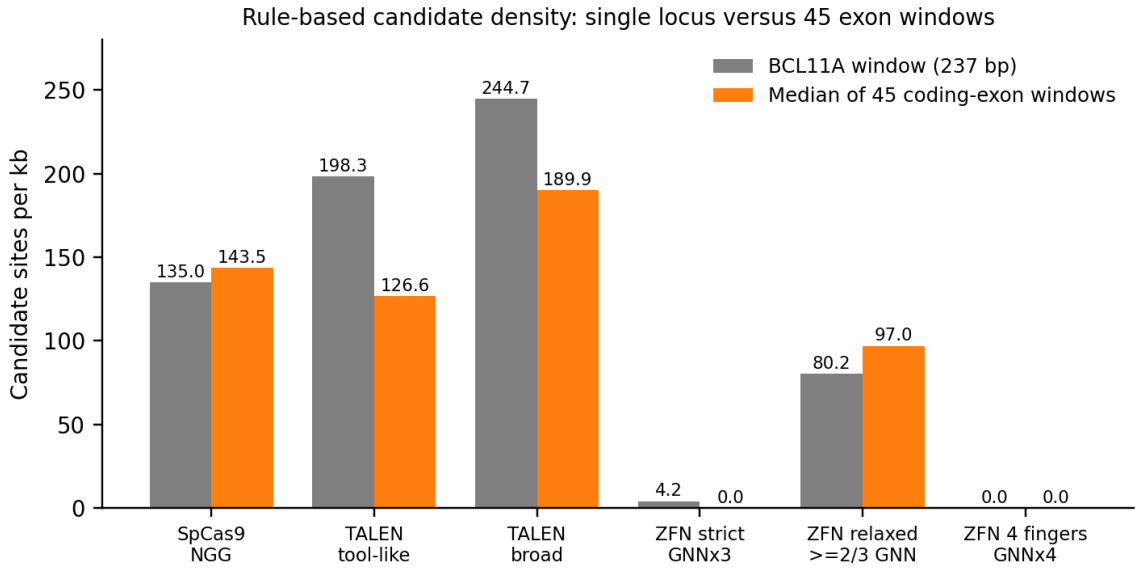


Figure 20. Candidate sites per kb under the rules of Section 2.6: BCL11A window (grey) versus the median of 45 coding-exon windows (orange). Plotted from the values in Table 12; rule-based counts, not tool outputs. Windows are drawn only from exons of at least 237 bp, so the sample favours longer exons.

3.6 What was obtained for each platform

Table 13. Results of this study (all computational). The footprint row reports spans as they are; it is not a complexity measure, because the spacer is included for two platforms only.

Parameter	SpCas9	TALEN	ZFN
Recognition unit	~20-nt guide + NGG PAM	One RVD per base; 5' T0	One finger per ~3 bp
Bases specified for this target	20	34 (arms)	18 (2 x 9)
Parts to assemble	1 spacer (Cas9 fixed)	34-36 RVDs, 2 proteins (full 18 + 18 read from the tool table)	~6 fingers, 2 proteins
Tool score	76.99 (CHOPCHOP); Doench '16 = 77	Not reported	Not obtained
Repair model	83.91% frameshift outcomes (Cas9 model)	Not modelled	Not modelled
Off-target output	None at 0-2 mismatches; CHOPCHOP 1 and CRISPOR 4 at 3	Selected for 0 off-target pairs; half-site MM2 = 15/26 as listed	Not assessed
Footprint	23 bp incl. PAM	51 bp incl. spacer	25 bp incl. spacer
Tool status	CHOPCHOP + CRISPOR	CHOPCHOP TALEN mode	None; rule-based scan
Site density, BCL11A / median of 45 windows (per kb)	135.0 / 143.5	198.3 or 244.7 / 126.6 or 189.9 (two rules)	4.2 / 0.0 (strict); 80.2 / 97.0 (relaxed)

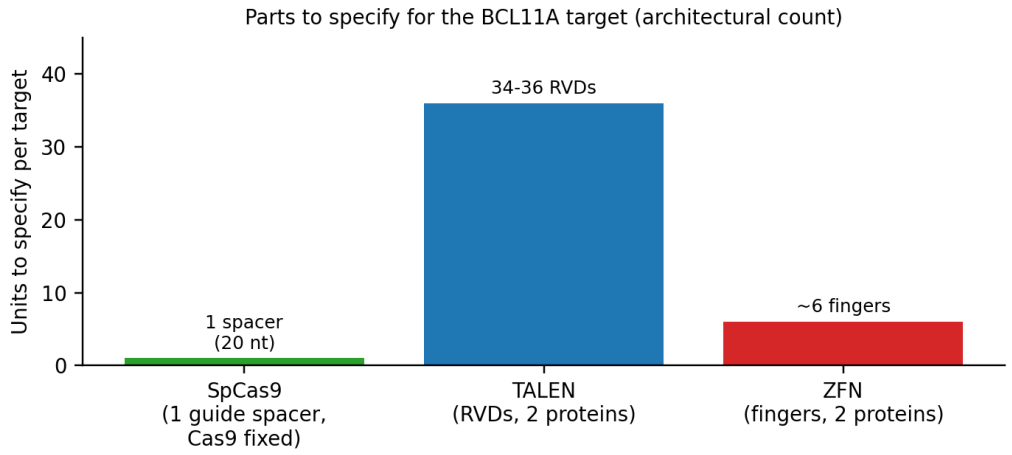


Figure 3. Units that must be specified for the BCL11A target: one guide spacer for SpCas9 (the Cas9 protein is fixed), 34-36 RVDs in two proteins for the TALEN pair and about six fingers in two proteins for the ZFN pair. This is an architectural count, not a measure of construction effort.

4. Discussion

4.1 What the data support and what they do not

Table 14. Status of each possible claim after the single-locus and 45-window analyses.

Claim	Status	Basis
Cas9 specifies the fewest units	Architectural fact, not a locus result	One spacer versus 34-36 RVDs or ~6 fingers (Figure 3); holds for any target
Fewer units means less effort	Not shown	ZFN specifies fewer bases (18) than Cas9 (20) yet is harder to build [29,30]; units are not construction effort
Cas9 has more candidate sites	Not supported	Single window: units differ and TALEN rules are permissive. 45 windows: Cas9 143.5 per kb versus TALEN 126.6 or 189.9 depending on the rule. Strict ZFN sites are rare (median 0.0 per kb)
Predicted efficiency	Not comparable	Only Cas9 has a model score

Claim	Status	Basis
Specificity	Not decided	Different units across platforms; ZFN not assessed
Cost, time, multiplexing	Literature only	Not measured here

4.2 Structural insights: protein-DNA interface versus RNA-DNA hybridisation

Every protein-based nuclease couples recognition to a protein fold. ZFNs inherit the finger-to-finger coupling of Cys2His2 arrays, so the unit that is easy to describe (a triplet) is not the unit that is easy to build. TALEs solved the coupling problem with a near one-to-one code, at the price of long repetitive coding sequences that recombine and are hard to amplify [8,10,26]. In both cases a new protein must be engineered, cloned, expressed and checked for every target. CRISPR-Cas9 relocates recognition to the RNA: the Cas9 protein is a fixed machine that reads a PAM, and the spacer supplies the address by base pairing [11,12,15]. In this case study the CRISPR specification is the single string CCTGAAGGGATACCAACCCG; the TALEN specification is the RVD arrays of Table 9 plus scaffold, linker and spacer choices; and the ZFN specification is a pair of 3-finger arrays whose performance cannot be read from sequence alone (Figure 3).

Because Watson-Crick energetics are uniform across targets, CRISPR scoring models (Doench '16, MIT, CFD, Lindel, Shen et al.) can be trained once and applied to any guide [18,19,20,27], although they are only modestly predictive. No equally general score exists for ZFN context effects, and TALEN binding prediction remains noisy: algorithms agree only partly, RVDs differ in affinity and neighbour effects are not captured [25]. In silico, a TALEN design is also only a few clicks in CHOPCHOP, so the difference lies mainly in construction and validation, which this study did not measure. The 45-window run adds a further point: the PAM (NGG) and GNN rules both depend on base composition, so the number of usable sites is a property of the sequence as well as of the platform.

4.3 Time to experiment (literature)

Time-to-experiment differences come from the build step, not from the in-silico design. The published modular-assembly protocol for ZFNs is described as taking about 24-26 days [2] [verify]. TALEN assembly needs multi-step cloning of repeat arrays: the Cermak et al. Golden Gate system takes about 5 days, with a correct clone typically found by screening 3 white colonies [8], and the Sanjana toolbox delivers TALENs within a week [10]; before such methods the repetitive coding sequence was unstable in ordinary PCR and expensive to synthesise. For CRISPR-Cas9, a guide is ordered as a short synthetic oligonucleotide and used with a fixed Cas9, so the build step can be reduced to days [add a source for the time estimate, since '24 hours' depends on supplier and format]. These timings were not measured in this project.

4.4 Multiplexing, cost and efficiency (literature)

Because only the spacer changes, several guides can be expressed with one Cas9, and multiplexing was shown in the first human-cell Cas9 papers [13]. TALENs have achieved multi-target goals, for example all three MLO homoeoalleles in hexaploid wheat [44], although with much more cloning effort per target [8,10]. Editing efficiency is locus- and cell-dependent: for TALENs, about 14-17% modified alleles at CCR5 and IL2RG were reported in a head-to-head comparison [9], and the corresponding ZFN and Cas9 values are not quoted here because they were not in the supplied sources. Cost and time were not

measured. CRISPR multiplexing carries risks of added off-target sites, translocations between simultaneous breaks and p53-mediated selection [40,49]. Multiplying parts by the number of genes (Table 15) is arithmetic, not evidence of performance.

Table 15. Components to specify for n independent knockouts, extrapolated from the single-gene designs (derived illustration only).

Genes to knock out (n)	CRISPR-Cas9: guides (one Cas9)	TALEN: RVDs (proteins)	ZFN: fingers (proteins)
1	1	36 (2)	~6 (2)
5	5	180 (10)	~30 (10)
10	10	360 (20)	~60 (20)

4.5 Where protein-based nucleases remain competitive

TALENs have no PAM restriction beyond the 5' T, a long footprint and a favourable cytotoxicity profile relative to ZFNs. They enable protein-only delivery, for example to organelles: mitochondria-targeted TALENs cured cytoplasmic male sterility in rice and rapeseed [45], and TALE arrays have found a second life as the DNA-binding module of base editors for organellar genomes [47]. TALEN-edited universal CAR T cells produced molecular remission in infants with B-ALL [35]. Our TALEN pair was selected for 0 predicted off-target pairs, and pairing does give TALENs a specificity mechanism, but it is not shown here to be better than a single Cas9 guide. CRISPR-Cas9 has its own limits: the PAM requirement, tolerance of distal mismatches (1 and 4 sites at 3 mismatches in CHOPCHOP and CRISPOR, 44 at 4), immunity to bacterial Cas proteins [48], AAV packaging size, and DSB-associated risks including large deletions, translocations and p53 activation [40,41,49]. High-fidelity variants (SpCas9-HF1, eSpCas9(1.1), HypaCas9) and paired nickases reduce off-target activity [28,42,52], and CRISPOR supports re-scoring the same guide for them.

4.6 Context: BCL11A and the therapeutic enhancer strategy

BCL11A encodes a repressor of fetal haemoglobin (gamma-globin) [23]. In sickle cell disease and beta-thalassaemia, disrupting the erythroid-specific enhancer of *BCL11A* reactivates fetal haemoglobin, the strategy behind exagamglogene autotemcel (Casgevy) [21,22,24]. Targeting the enhancer rather than the coding sequence restricts the loss of BCL11A to the erythroid lineage. This study deliberately uses a coding exon because it gives a knock-out readout (frameshift prediction) that is comparable in design terms across platforms; it is a methodological demonstration, not a therapeutic design.

4.7 Limitations

- **One benchmark locus plus a sampled extension.** Platform properties cannot be separated from properties of the BCL11A exon. The 45-window extension used only exons of at least 237 bp (23,754 of 216,361, about 11%), so the sample favours longer exons, and windows are one per exon, so the medians describe coding-exon windows, not the genome.
- **Selection asymmetry.** The Cas9 guide is the gene-wide Rank 1 and defined the window; the TALEN pair is Rank 3, restricted to the same exon and selected for 0 off-target pairs. The three designs are at different positions.
- **Site counts are descriptive.** Units differ, the TALEN rules are permissive, no tool filters were applied, and only Cas9 was validated against a tool. Counts depend on base composition.

- **ZFN arm is rule-based.** One strict rule, no tool output and no specificity assessment; it cannot support a three-platform ranking. Finger numbering depends on binding direction and should be checked before finger names are assigned.
- **TALEN items still open.** The T0/terminal-A reading, the orientation of the right-hand array, the meaning of '15/26', and the depth of mismatch searched by the tool.
- **Cas9 repair model.** It is specific to Cas9 blunt-end breaks and does not describe TALEN or ZFN outcomes. Frameshift in a coding exon is a proxy for knock-out; exon coding status and isoform coverage were not confirmed.
- **Tool dependence.** Both CHOPCHOP arms share tool conventions, the two Cas9 efficiency scores are probably the same model, and the 3-mismatch discrepancy (1 versus 4) is unexplained.
- **Multi-locus statistics.** Medians only; no interquartile range or test was applied, and the run used one random seed.
- **References.** Entries marked with a dagger were added from general knowledge and must be checked against the journal pages.

4.8 Validation roadmap

Table 16. Suggested wet-lab and computational validation of the predictions.

Question	Recommended assay	Applied to
Does the nuclease cut at the intended site?	T7E1 (or Surveyor) mismatch-cleavage assay on the 236-bp Pair 1 amplicon	All three designs
What indels and frame distribution result?	Amplicon next-generation sequencing; compare with the predicted 83.91% frameshift, +1/+2 split and 77.42% microhomology deletions	CRISPR guide; also TALEN and ZFN
Are predicted off-target sites cut?	Targeted deep sequencing at chr5:82019991 and the four 3-mismatch sites; unbiased GUIDE-seq [31], CIRCLE-seq [50] or Digenome-seq [51]	CRISPR (and TALEN look-alike sites)
Is the TALEN pair specific and non-toxic?	Targeted sequencing at pair and half-site look-alikes; cell survival and gamma-H2AX/53BP1 foci assays	TALEN
Is the ZFN candidate valid?	Re-run with ZiFiT or ZFNGenome when reachable [3,4]; assemble fingers and test binding and cleavage [2,29]	ZFN
Does the multi-locus pattern hold?	Repeat the count over more windows and seeds; compare TALEN counts with CHOPCHOP's full lists	All rules
Is the in-frame fraction a problem?	Clonal sequencing; Western blot or functional assay for residual protein	All

4.9 Conclusions

At one coding exon of human *BCL11A*, CRISPR-Cas9 reduced the specification to one 20-nt spacer beside an NGG PAM, with a predicted efficiency score of 76.99 (Doench '16 = 77, probably the same model), no predicted look-alikes at 0-2 mismatches in either tool and a Cas9-model prediction of 83.91% frameshift outcomes. A CHOPCHOP TALEN pair in the same exon needed two arrays of 18 RVDs each (34-36 repeats to specify), and a rule-based scan gave one candidate ZFN pair under a strict GNNx3 rule. The smaller number of units for Cas9 follows from the architecture of the platforms and does not depend on the locus. Candidate-site counts differ with the rules used. Across 45 coding-exon windows, Cas9 and TALEN candidate densities were of the same order (medians 143.5 versus 126.6 or 189.9 per kb, depending on the TALEN rule) and strict GNNx3 ZFN pairs were absent from the median window (0.0 per kb), so the data do not show that Cas9 has more target sites, but they do

show how restrictive strictly modular ZFN design is. The study does not test efficiency, specificity, cost, multiplexing or time to experiment. Those claims rest on the literature, and wet-lab validation and a larger multi-locus survey are the necessary next steps.

5. Consolidated Summary of Results

This section gathers every result in one place. All values are predicted or rule-based; none is experimental.

Table 17. All results in one table.

Area	Result	Source
Locus	BCL11A coding exon region, chr2:60,462,161-60,462,397 (237 nt), hg38; three designs within ~30 bp	UCSC, CHOPCHOP
Cas9 guide (Rank 1)	CCTGAAGGGATACCAACCCG + CGG, chr2:60462265 (+), GC 60%, self-compl. 0	CHOPCHOP (Table 5)
Predicted efficiency	76.99; Doench '16 = 77; Moreno-Mateos 64; Rule Set 3 = 81 (probably not independent)	CHOPCHOP, CRISPOR
Cas9 off-targets	0/0/0 at 0-2 mismatches; 3 mm: 1 (CHOPCHOP) vs 4 (CRISPOR); 4 mm: 44; MIT 94, CFD 98	CHOPCHOP, CRISPOR
Cas9 repair model	Frameshift 83.91% (+1 47.58%, +2 36.33%); in-frame 16.09%; microhomology deletions 77.42%; precision 0.47	Shen et al. via CHOPCHOP
Primers	Pair 1, 236 bp, Tm 60.2 C (TACAACTTGCAAACAGCCATTC / CCTCTCTCGATACTGATCCTGG)	CHOPCHOP
TALEN pair (Rank 3)	T0 (60,462,267) + 17 + 15-bp spacer + 17 + terminal A; 51-bp footprint; selected for 0 off-target pairs; half-site MM2 15/26 as listed; 18 + 18 RVDs listed	CHOPCHOP TALEN mode
ZFN candidate	GTGGCTGGG - AAGGACA (7 bp) - GGTGCAGAA, chr2:60462236-60462260; 3 + 3 fingers; not scored	Rule-based scan
Single-window counts	Cas9 32; TALEN 47 (tool-like), 58 (broad); ZFN 1 (strict), 19 (relaxed), 0 (4 fingers); 18 Cas9 guides in the 120-bp CRISPOR window (matches CRISPOR)	Script S1
Multi-locus run	216,361 unique coding exons; 23,754 of at least 237 bp; 500 sampled (seed 1); 45 windows used	make_exons.py, fetch_windows.py
Median density, 45 windows (per kb)	Cas9 143.5; TALEN 126.6 (tool-like), 189.9 (broad); ZFN strict 0.0, relaxed 97.0, 4 fingers 0.0	Script S1 (Table 12)
Verdict	Design architecture favours Cas9 (fewest units). Site availability not higher for Cas9 than TALEN; strict ZFN sites rare. Efficiency, specificity, cost, multiplexing and time not tested here	Section 4.1

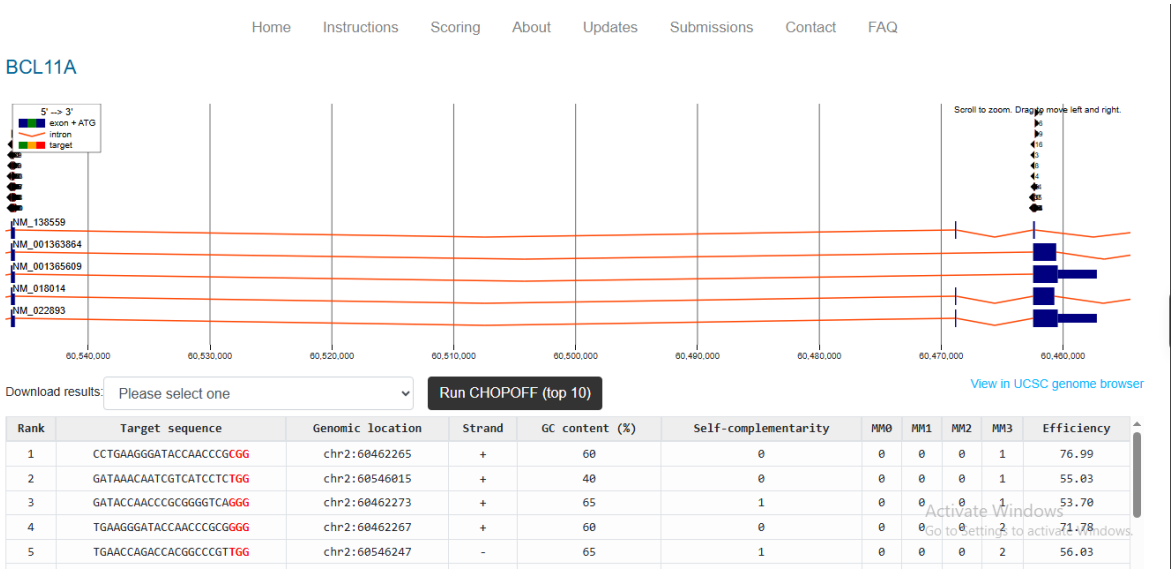
Table 18. Review checklist: what was done and what remains open.

Item	Status
Narrowed thesis, abstract and conclusion	Done
Table of values from this study separated from literature values	Done
Candidate-site count for the three platforms (single locus)	Done (Section 3.4, script S1, Figure 22)
Candidate-site count across exon windows	Done: 45 windows (Section 3.5); medians to be confirmed against results.csv
Window rule for exons longer than 237 bp	Done: central 237 bp
Full 18-repeat TALEN arrays	Read from the tool table (Supplementary Figure S2); retake as a clean screenshot
ZFN design tool (ZiFiT / ZFNGenome)	Not reached; retry from another network

Item	Status
TALEN T0 reading, right-array orientation, meaning of '15/26'	Open: confirm in CHOPCHOP documentation or TALE-NT
Interquartile range and share of zero-strict-ZFN windows	Open: compute from results.csv
TALEN counts compared with CHOPCHOP's full list for the window	Open
References marked with a dagger	Open: verify against journal pages (reference [4] ZFNGenome, BMC Genomics 2011;12:83, was checked and is correct)
Watermark on screenshots (Figures 1-18)	Open: recapture without the 'Activate Windows' text
Authors, affiliation, AI-use statement, Python version	Open: placeholders

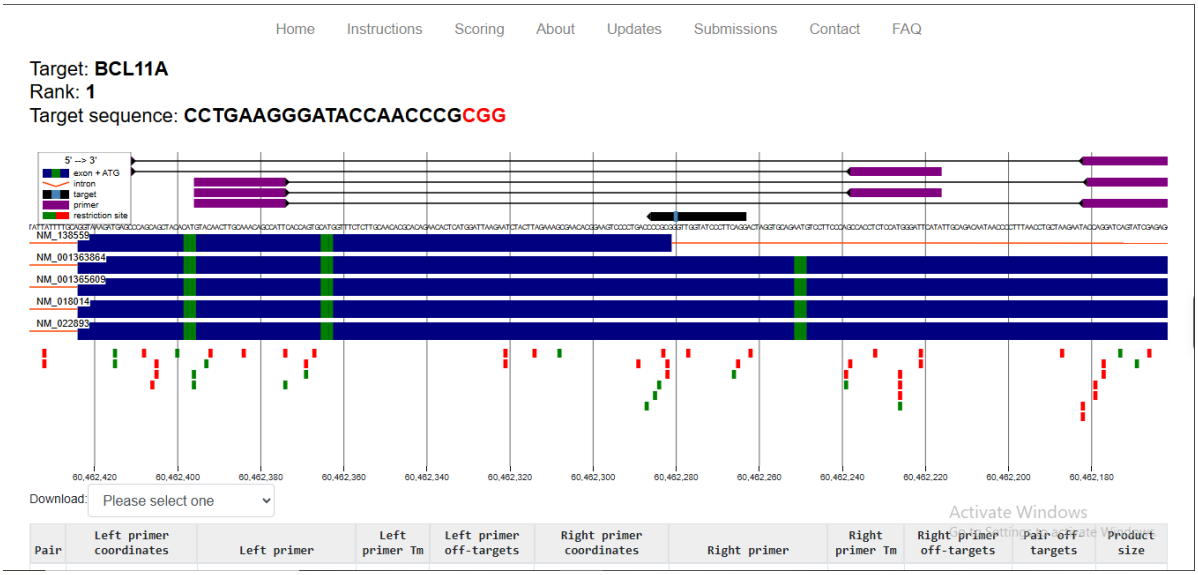
6. Figures: Tool Screenshots and Script Output

Figures 1-3, 19 and 20 are schematics or charts generated for this paper. Figures 4-18, 21 and 22 are unmodified screenshots with the source tool noted. Screenshots carry a faint operating-system 'Activate Windows' watermark that was left untouched. Unless stated otherwise: genome build GRCh38/hg38; tools accessed 24 October 2026; numerical values are predicted.



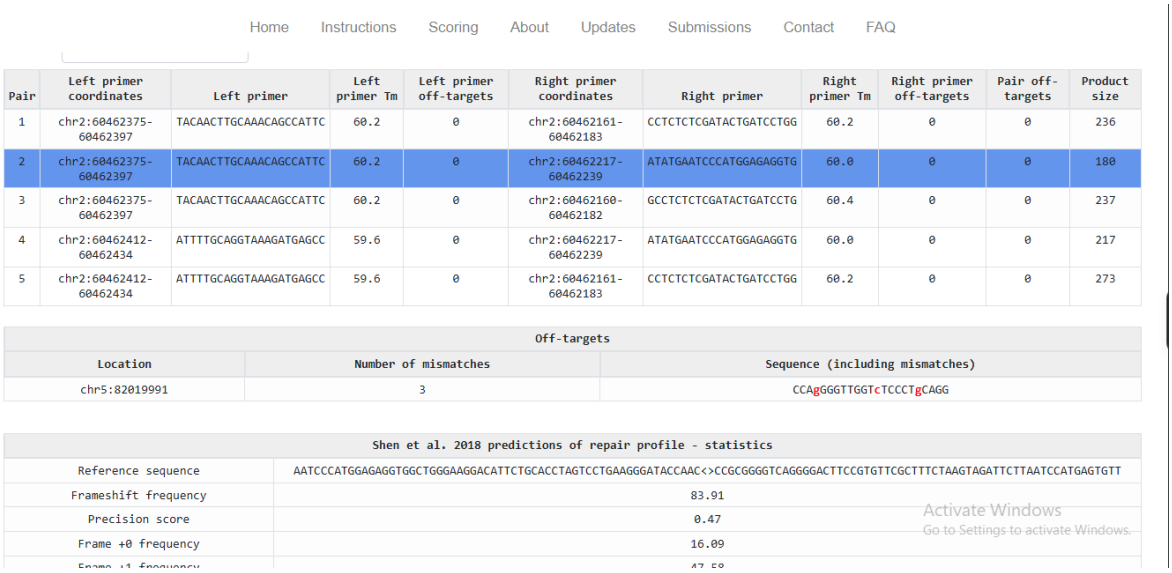
Rank	Target sequence	Genomic location	Strand	GC content (%)	Self-complementarity	MM0	MM1	MM2	MM3	Efficiency
1	CCTGAAGGGATACCAACCCGCGG	chr2:60462265	+	60	0	0	0	0	1	76.99
2	GATAAACAATCGTCATCCTCTGG	chr2:60546015	+	40	0	0	0	0	1	55.03
3	GATACCAACCCGCGGGGTCAGGG	chr2:60462273	+	65	1	0	0	0	1	53.70
4	TGAAGGGATACCAACCCGCGGGG	chr2:60462267	+	60	0	0	0	0	2	71.78
5	TGAACCAGACCACGGCCCGTTGG	chr2:60546247	-	65	1	0	0	0	2	56.03

Figure 4. CHOPCHOP results for BCL11A (hg38, CRISPR/Cas9, knock-out). Gene map above the ranked guide table. Rank 1: CCTGAAGGGATACCAACCCG (PAM CGG), chr2:60462265 (+), predicted efficiency 76.99, GC 60%, no look-alikes at 0-2 mismatches.



Pair	Left primer coordinates	Left primer	Left primer Tm	Left primer off-targets	Right primer coordinates	Right primer	Right primer Tm	Right primer off-targets	Pair off-targets	Product size

Figure 5. Rank 1 guide inside the BCL11A gene model. Black bar = guide; blue mark = approximate predicted cut, about 3 nt 5' of the PAM; dark-blue bands = exons; purple bars = suggested primers; green and red marks = restriction sites.

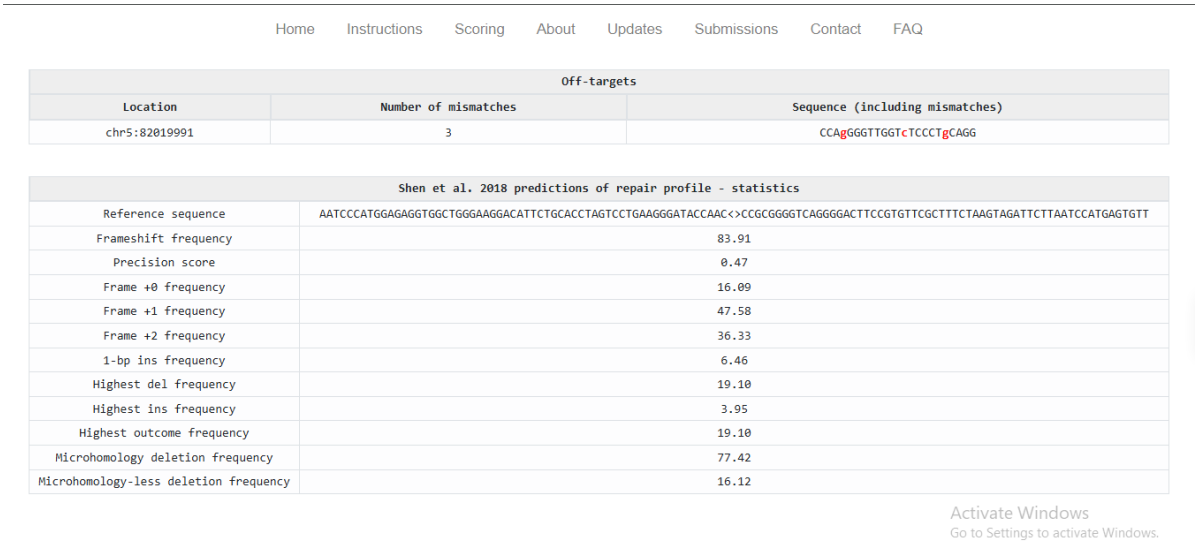


Pair	Left primer coordinates	Left primer	Left primer Tm	Left primer off-targets	Right primer coordinates	Right primer	Right primer Tm	Right primer off-targets	Pair off-targets	Product size
1	chr2:60462375-60462397	TACAACTTGCAAACAGCCATTC	60.2	0	chr2:60462161-60462183	CCTCTCTCGATACTGATCCTGG	60.2	0	0	236
2	chr2:60462375-60462397	TACAACTTGCAAACAGCCATTC	60.2	0	chr2:60462217-60462239	ATATGAATCCCATGGAGAGGTG	60.0	0	0	180
3	chr2:60462375-60462397	TACAACTTGCAAACAGCCATTC	60.2	0	chr2:60462160-60462182	GCCTCTCTCGATACTGATCCTG	60.4	0	0	237
4	chr2:60462412-60462434	ATTTTGCAGGTAAAGATGAGCC	59.6	0	chr2:60462217-60462239	ATATGAATCCCATGGAGAGGTG	60.0	0	0	217
5	chr2:60462412-60462434	ATTTTGCAGGTAAAGATGAGCC	59.6	0	chr2:60462161-60462183	CCTCTCTCGATACTGATCCTGG	60.2	0	0	273

Off-targets		
Location	Number of mismatches	Sequence (including mismatches)
chr5:82019991	3	CCAgGGGTTGGTcTCCCTgCAGG

Shen et al. 2018 predictions of repair profile - statistics	
Reference sequence	AATCCCATGGAGAGGTGGCTGGGAAGGACATTCTGCACCTAGTCCTGAAGGGATACCAAC<>CCGCGGGGTCAGGGGACTTCCGTGTTCGCTTTCTAAGTAGATTCTTAATCCATGAGTGTT
Frameshift frequency	83.91
Precision score	0.47
Frame +0 frequency	16.09

Figure 6. Predicted primer pairs and the single predicted off-target site for the Rank 1 guide. Pair 1 (TACAACTTGCAAACAGCCATTC / CCTCTCTCGATACTGATCCTGG; Tm 60.2 C; 236 bp) was selected. The off-target box shows chr5:82019991 with 3 mismatches (lowercase red).



Off-targets		
Location	Number of mismatches	Sequence (including mismatches)
chr5:82019991	3	CCAgGGGTTGGTcTCCCTgCAGG

Shen et al. 2018 predictions of repair profile - statistics	
Reference sequence	AATCCCATGGAGAGGTGGCTGGGAAGGACATTCTGCACCTAGTCCTGAAGGGATACCAAC<>CCGCGGGGTCAGGGGACTTCCGTGTTCGCTTTCTAAGTAGATTCTTAATCCATGAGTGTT
Frameshift frequency	83.91
Precision score	0.47
Frame +0 frequency	16.09
Frame +1 frequency	47.58
Frame +2 frequency	36.33
1-bp ins frequency	6.46
Highest del frequency	19.10
Highest ins frequency	3.95
Highest outcome frequency	19.10
Microhomology deletion frequency	77.42
Microhomology-less deletion frequency	16.12

Figure 7. Predicted repair outcomes after Cas9 cuts at the Rank 1 site (Shen et al. 2018 model). Frameshift 83.91%; precision 0.47; +0 16.09%, +1 47.58%, +2 36.33%; microhomology deletions 77.42%. Model predictions, not laboratory results.

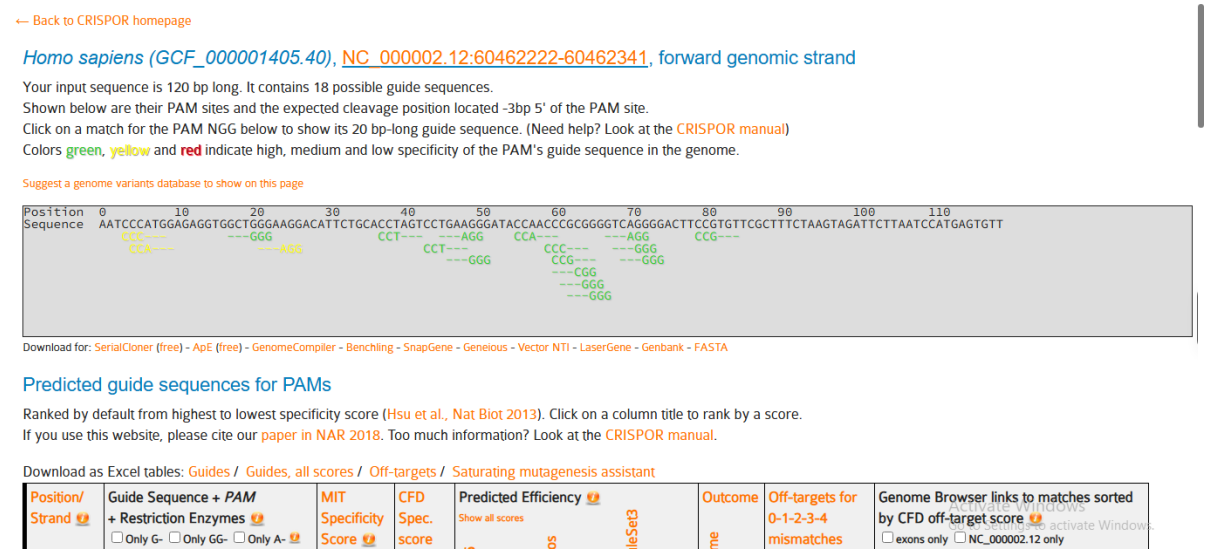
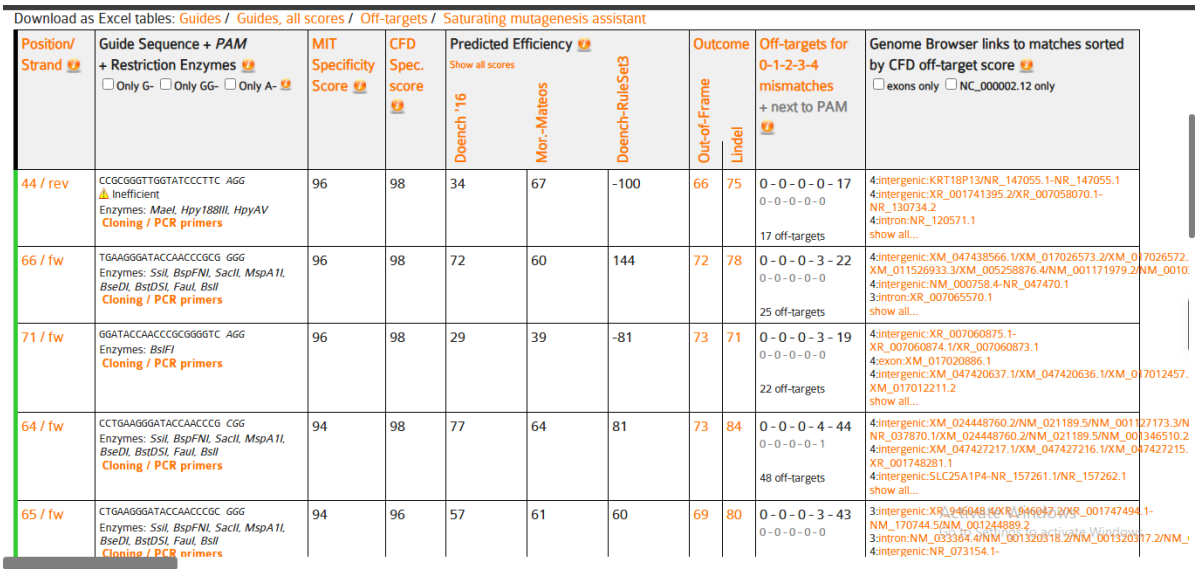


Figure 8. CRISPOR map of the 120-bp input (hg38). CRISPOR found 18 possible guides. The header region NC_000002.12:60462222-60462341 is hg38 and the Rank 1 guide starts at 60462265, matching CHOPCHOP. (The CRISPOR input-page screenshot of the first draft showed hg19 and a stray marker, so it is omitted; this header confirms the build.)



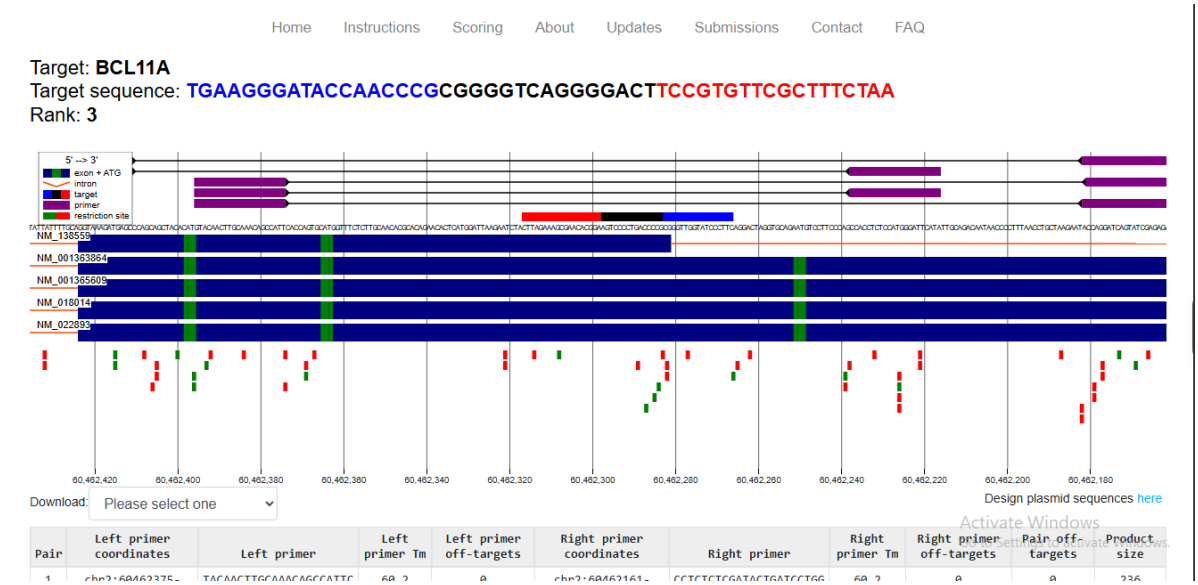
Position/ Strand	Guide Sequence + PAM + Restriction Enzymes	MIT Specificity Score	CFD Spec. score	Predicted Efficiency (Doench '16)	Predicted Efficiency (Mor.-Mateos)	Predicted Efficiency (Doench-RuleSet3)	Outcome (Out-of-Frame)	Outcome (Lindel)	Off-targets for 0-1-2-3-4 mismatches + next to PAM	Genome Browser links to matches sorted by CFD off-target score
44 / rev	CCGCGGGTTGGTATCCCTTC AGG ⚠ Inefficient Enzymes: *MaeI, Hpy188III, HpyAV* Cloning / PCR primers	96	98	34	67	-100	66	75	0 - 0 - 0 - 0 - 17 0 - 0 - 0 - 0 - 0 17 off-targets	4:intergenic:KRT18P13/NR_147055.1-NR_147055.1 4:intergenic:XR_001741395.2/XR_007058070.1-NR_130734.2 4:intron:NR_120571.1 show all...
66 / fw	TGAAGGGATACCAACCCGCG GGG Enzymes: *SsiI, BspFNI, SacII, MspA1I, BseDI, BstDSI, FauI, BsiI* Cloning / PCR primers	96	98	72	60	144	72	78	0 - 0 - 0 - 3 - 22 0 - 0 - 0 - 0 - 0 25 off-targets	4:intergenic:XM_047438566.1/XM_017026573.2/XM_017026572.-XM_011526933.3/XM_005258876.4/NM_001171979.2/NM_0010-4:intergenic:NM_000758.4-NR_047470.1 3:intron:XR_007065570.1 show all...
71 / fw	GGATACCAACCCGCGGGGTC AGG Enzymes: *BsiFI* Cloning / PCR primers	96	98	29	39	-81	73	71	0 - 0 - 0 - 3 - 19 0 - 0 - 0 - 0 - 0 22 off-targets	4:intergenic:XR_007060875.1-XR_007060874.1/XR_007060873.1 4:exon:XM_017020886.1 4:intergenic:XM_047420637.1/XM_047420636.1/XM_017012457.-XM_017012211.2 show all...
64 / fw	CCTGAAGGGATACCAACCCG CGG Enzymes: *SsiI, BspFNI, SacII, MspA1I, BseDI, BstDSI, FauI, BsiI* Cloning / PCR primers	94	98	77	64	81	73	84	0 - 0 - 0 - 4 - 44 0 - 0 - 0 - 0 - 1 48 off-targets	4:intergenic:XM_024448760.2/NM_021189.5/NM_001127173.3/N-NR_037870.1/XM_024448760.2/NM_021189.5/NM_001346510.2 4:intergenic:XM_047427217.1/XM_047427216.1/XM_047427215.-XR_001748281.1 4:intergenic:SLC25A1P4-NR_157261.1/NR_157262.1 show all...
65 / fw	CTGAAGGGATACCAACCCGC GGG Enzymes: *SsiI, BspFNI, SacII, MspA1I, BseDI, BstDSI, FauI, BsiI* Cloning / PCR primers	94	96	57	61	60	69	80	0 - 0 - 0 - 3 - 43 0 - 0 - 0 - 0 - 0	3:intergenic:XR_946048.4/XR_946047.2/XR_001747494.1-NM_170744.5/NM_001244889.2 3:intron:NM_033364.4/NM_001320318.2/NM_001320317.2/NM_ 4:intergenic:NR_073154.1-

Figure 9. CRISPOR scores for candidate guides (Rank 1 guide = fourth row, 64/fw). MIT 94, CFD 98, Doench '16 = 77 (Moreno-Mateos 64, Rule Set 3 = 81), out-of-frame 73, Lindel 84; off-targets 0-0-0-4-44 (48 in total). The neighbouring guide (66/fw) has MIT 96 but lower predicted efficiency (72).



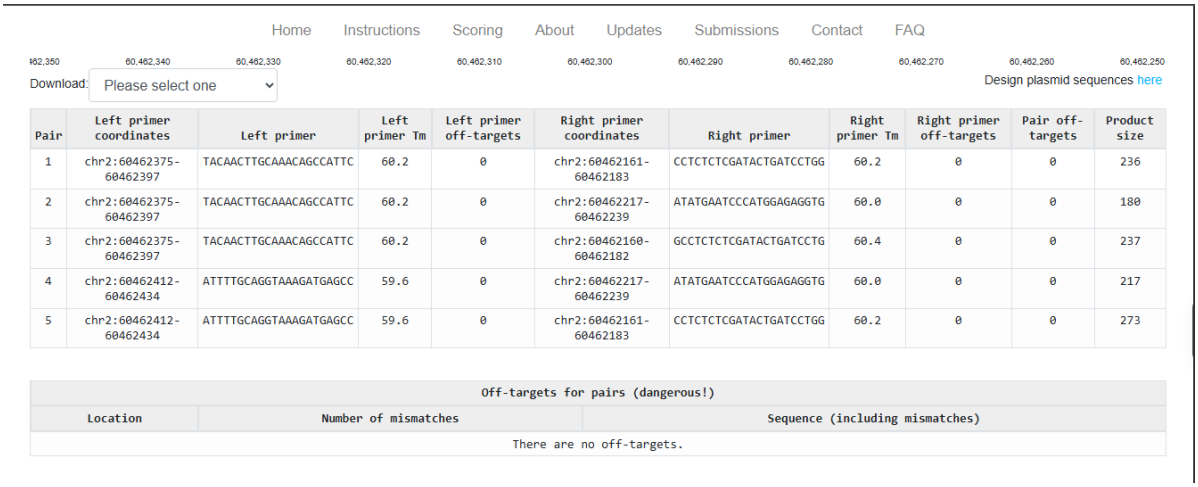
Rank	Target sequence	Genomic location	TALE 1	TALE 2	Cluster	Off-target pairs	Off-targets MM0	Off-targets MM1	Off-targets MM2	Off-targets MM3	Restriction sites	Best ID
1	TTACGCGAGAATTCCCGTTTGCTTAAGTGCTGGGGTTTGCCTTGCTTGCGGCGA	chr2:60553212	NG NG NI HD	NG HD NH HD	114	0	0/0	0/0	2/13	0/0	BseYI:12 SmlI:5 AflII:5	1

Figure 10. CHOPCHOP TALEN-mode results for BCL11A (hg38). Two clusters of candidate pairs (near chr2:60,553,000 and chr2:60,462,000). Rank 1 (chr2:60553212) lies in a different exon from the CRISPR guide and was not used.



Pair	Left primer coordinates	Left primer	Left primer Tm	Left primer off-targets	Right primer coordinates	Right primer	Right primer Tm	Right primer off-targets	Pair off-targets	Product size
1	chr2:60462375-	TACAACTTGCAAACAGCCATTC	60.2	0	chr2:60462161-	CCTCTCTCGATACTGATCCTGG	60.2	0	0	236

Figure 11. Detail view of the selected TALEN pair (Rank 3, chr2:60462267). Blue = left part of the target string, black = spacer (15 bp, CGGGGTCAGGGGACT), red = right part. The coordinate axis runs from high to low, so left and right appear flipped; this is a display direction.



Home Instructions Scoring About Updates Submissions Contact FAQ

62,350 60,462,340 60,462,330 60,462,320 60,462,310 60,462,300 60,462,290 60,462,280 60,462,270 60,462,260 60,462,250

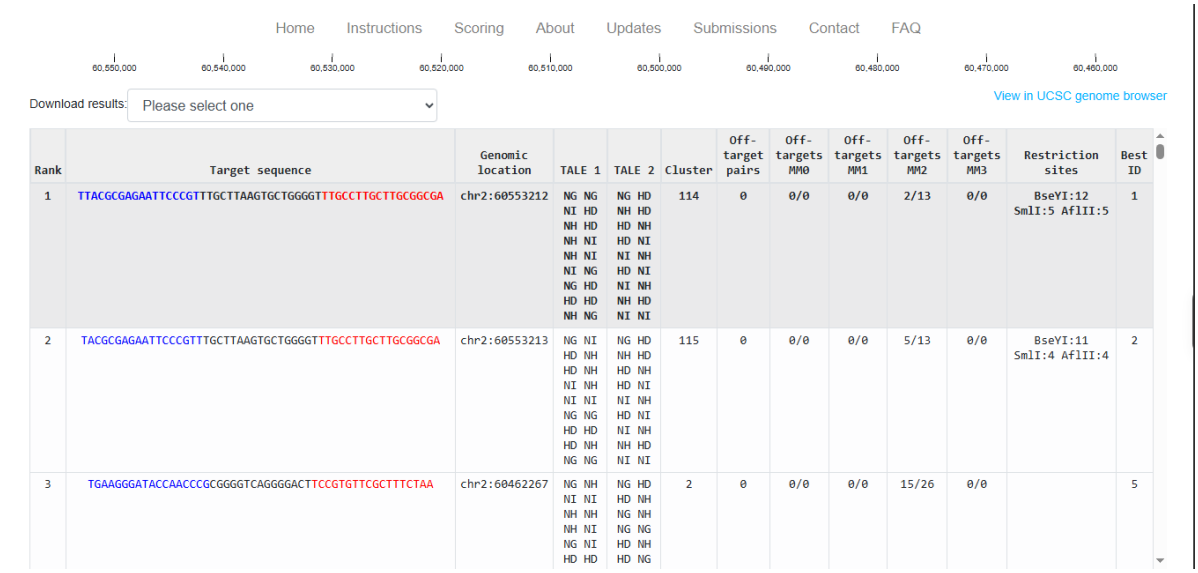
Download: Please select one

Design plasmid sequences here

Pair	Left primer coordinates	Left primer	Left primer Tm	Left primer off-targets	Right primer coordinates	Right primer	Right primer Tm	Right primer off-targets	Pair off-targets	Product size
1	chr2:60462375-60462397	TACAACTTGCAAACAGCCATTC	60.2	0	chr2:60462161-60462183	CCTCTCTCGATACTGATCCTGG	60.2	0	0	236
2	chr2:60462375-60462397	TACAACTTGCAAACAGCCATTC	60.2	0	chr2:60462217-60462239	ATATGAATCCCATGGAGAGGTG	60.0	0	0	180
3	chr2:60462375-60462397	TACAACTTGCAAACAGCCATTC	60.2	0	chr2:60462160-60462182	GCCTCTCTCGATACTGATCCTG	60.4	0	0	237
4	chr2:60462412-60462434	ATTTTGCAGGTAAAGATGAGCC	59.6	0	chr2:60462217-60462239	ATATGAATCCCATGGAGAGGTG	60.0	0	0	217
5	chr2:60462412-60462434	ATTTTGCAGGTAAAGATGAGCC	59.6	0	chr2:60462161-60462183	CCTCTCTCGATACTGATCCTGG	60.2	0	0	273

Off-targets for pairs (dangerous!)		
Location	Number of mismatches	Sequence (including mismatches)
There are no off-targets.		

Figure 12. Predicted primer pairs for the Rank 3 target and the pair off-target check. Pair 1 as in Figure 6. The 'Off-targets for pairs' box reads 'There are no off-targets.'



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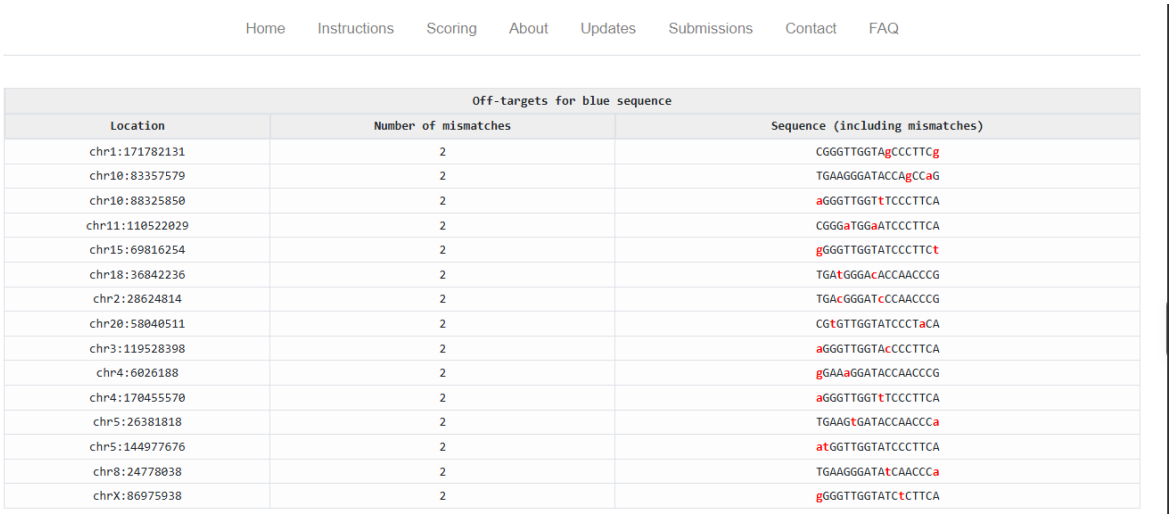
60,550,000 60,540,000 60,530,000 60,520,000 60,510,000 60,500,000 60,490,000 60,480,000 60,470,000 60,460,000

Download results: Please select one

View in UCSC genome browser

Rank	Target sequence	Genomic location	TALE 1	TALE 2	Cluster	Off-target pairs	Off-targets MM0	Off-targets MM1	Off-targets MM2	Off-targets MM3	Restriction sites	Best ID
1	TTACGCGAGAATTCCCGTTTGCTTAAGTGCTGGGGTTTGCCTTGCTTGCGGCGA	chr2:60553212	NG NG NI HD NH HD NH NI NH NI NI NG NG HD HD HD NH NG	NG HD NH HD HD NH HD NI NI NH HD NI NI NH NH HD NI NI	114	0	0/0	0/0	2/13	0/0	BseYI:12 SmlI:5 AflII:5	1
2	TACGCGAGAATTCCCGTTTGCTTAAGTGCTGGGGTTTGCCTTGCTTGCGGCGA	chr2:60553213	NG NI HD NH HD NH NI NH NI NI NG NG HD HD HD NH NG NG	NG HD NH HD HD NH HD NI NI NH HD NI NI NH NH HD NI NI	115	0	0/0	0/0	5/13	0/0	BseYI:11 SmlI:4 AflII:4	2
3	TGAAGGGATACCAACCCGCGGGGTCAGGGGACTTCCGTGTTCGCTTTCTAA	chr2:60462267	NG NH NI NI NH NH NH NI NG NI HD HD	NG HD HD NH NG NH NG NG HD NH HD NG	2	0	0/0	0/0	15/26	0/0		5

Figure 13. Rank 3 row of the TALEN results table with the RVD arrays. TALE 1 and TALE 2 list the RVDs of the left and right TALEN (first 12 per array visible here; all 18 are visible in Supplementary Figure S2). Rank 3: chr2:60462267, cluster 2, 0 off-target pairs, 0/0 at MM0 and MM1, 15/26 at MM2 (as reported), 0/0 at MM3.

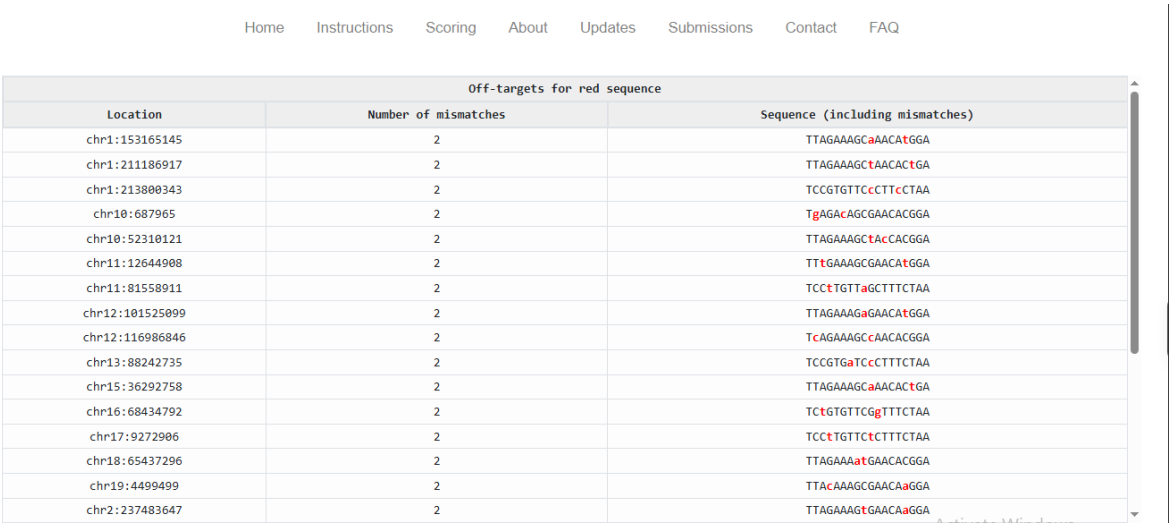


Home Instructions Scoring About Updates Submissions Contact FAQ

Off-targets for blue sequence

Location	Number of mismatches	Sequence (including mismatches)
chr1:171782131	2	CGGGTTGGTAgCCCTTCg
chr10:83357579	2	TGAAGGGATACCAgCCaG
chr10:88325850	2	aGGGTTGGTtTCCCTTCA
chr11:110522029	2	CGGGaTGGaATCCCTTCA
chr15:69816254	2	gGGGTTGGTATCCCTTCt
chr18:36842236	2	TGAtGGGAcACCAACCCG
chr2:28624814	2	TGAcGGGATcCCAACCCG
chr20:58040511	2	CGtGTTGGTATCCCTaCA
chr3:119528398	2	aGGGTTGGTAcCCCTTCA
chr4:6026188	2	gGAAaGGATACCAACCCG
chr4:170455570	2	aGGGTTGGTtTCCCTTCA
chr5:26381818	2	TGAAGtGATACCAACCCa
chr5:144977676	2	atGGTTGGTATCCCTTCA
chr8:24778038	2	TGAAGGGATAtCAACCCa
chrX:86975938	2	gGGGTTGGTATCtCTTCA

Figure 14. Two-mismatch look-alikes of the left TALEN half-site. Mismatched bases in lowercase red. A single look-alike cannot cause a cut because FokI needs both halves bound within the spacer range.



Home Instructions Scoring About Updates Submissions Contact FAQ

Off-targets for red sequence

Location	Number of mismatches	Sequence (including mismatches)
chr1:153165145	2	TTAGAAAGCaAACAtGGA
chr1:211186917	2	TTAGAAAGCtAACACtGA
chr1:213800343	2	TCCGTGTTCcCTTcCTAA
chr10:687965	2	TgAGAcAGCGAACACGGA
chr10:52310121	2	TTAGAAAGCtAcCACGGA
chr11:12644908	2	TTtGAAAGCGAACAtGGA
chr11:81558911	2	TCCtTGTTaGCTTTCTAA
chr12:101525099	2	TTAGAAAGaGAACAtGGA
chr12:116986846	2	TcAGAAAGCcAACACGGA
chr13:88242735	2	TCCGTGaTCcCTTTCTAA
chr15:36292758	2	TTAGAAAGCaAACACtGA
chr16:68434792	2	TCtGTGTTCGgTTTCTAA
chr17:9272906	2	TCCtTGTTCtCTTTCTAA
chr18:65437296	2	TTAGAAAatGAACACGGA
chr19:4499499	2	TTAcAAAGCGAACAaGGA
chr2:237483647	2	TTAGAAAGtGAACAaGGA

Figure 15. Two-mismatch look-alikes of the right TALEN half-site. As Figure 14 (the list scrolls beyond the rows shown).

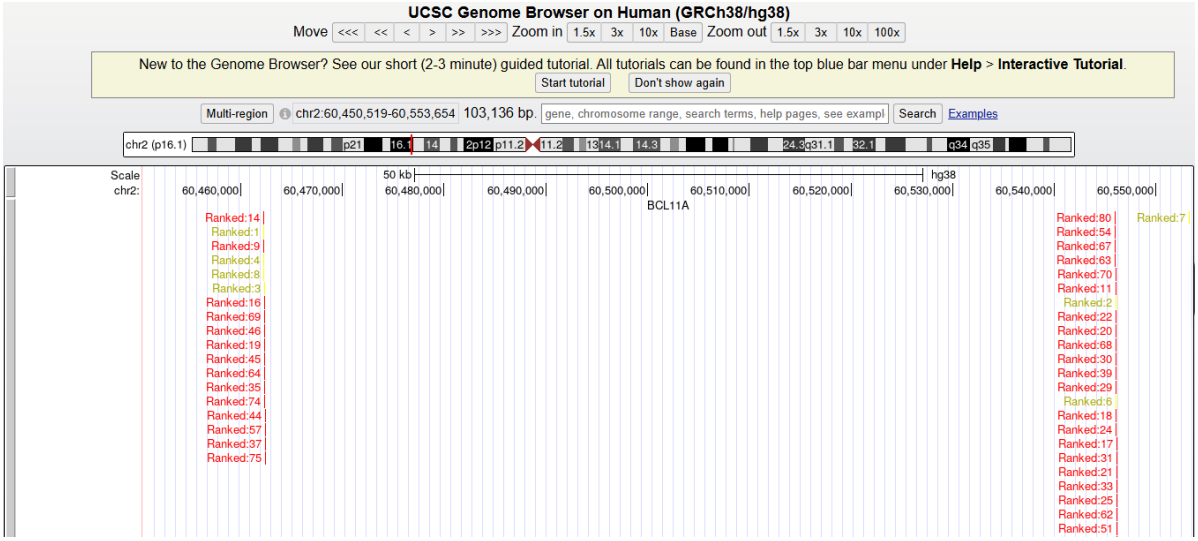


Figure 16. UCSC Genome Browser, whole BCL11A region (GRCh38/hg38): chr2:60,450,519-60,553,654 (103,136 bp) with the ranked CHOPCHOP target track; targets form two clusters near chr2:60,460,000 and chr2:60,545,000.

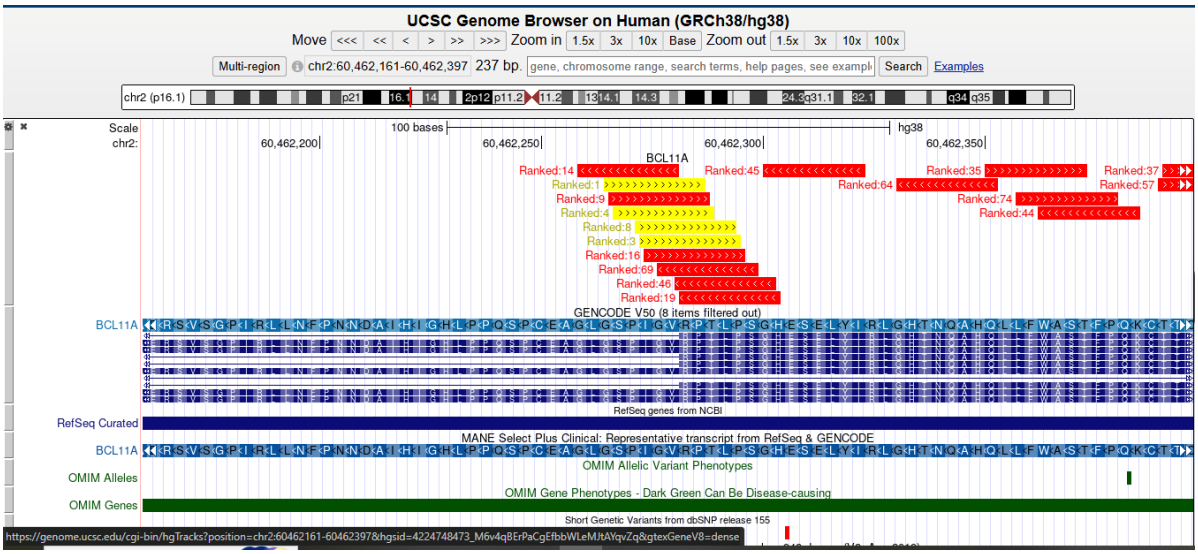


Figure 17. UCSC view of the 237-bp amplicon region chr2:60,462,161-60,462,397 (hg38). The position box shows exactly 237 bp; the right primer starts at 60462161 and the left primer ends at 60462397.



Figure 18. Sequence returned by the UCSC sequence service (JSON; hg38 chr2, start 60462160, end 60462397). The 'dna' field holds a 237-nt string beginning CCTCTCTCGATACTGATCCTGG and ending GTTGCAAGTTGTAC, in the same orientation as the CHOPCHOP results.

```
  1  CCTCTCTCGATACTGATCCTGGTATTCTTAGCAGGTTAAAGGGGTTATTGTCTGCAATAT

 61  GAATCCCATGGAGAGGTGGCTGGGAAGGACATTCTGCACCTAGTCCTGAAGGGATACCAA

121  CCCGCGGGGTCAGGGGACTTCCGTGTTCGCTTTCTAAGTAGATTCTTAATCCATGAGTGT

181  TCTGTGCGTGTTGCAAGAGAAACCATGCACTGGTGAATGGCTGTTTGCAAGTTGTAC
```

blue = ZFN left (76–84); grey = ZFN spacer (85–91); red = ZFN right half-site (92–100, read on the opposite strand)
green = Rank 1 CRISPR spacer (105–124); orange = PAM CGG (125–127); red letters = TALEN T0 (107) and terminal A (157); TALEN Rank 3 footprint = 107–157

Figure 19. Designs on the 237-nt amplicon sequence, numbered from 1 at chr2:60,462,161. Blue/grey/red = ZFN candidate (left half-site 76-84, 7-bp spacer 85-91, right half-site 92-100 read on the opposite strand as GGTGCAGAA); green = Rank 1 CRISPR spacer (105-124); orange = PAM CGG (125-127); red letters = TALEN T0 (107) and terminal A (157). The ZFN candidate is from a rule-based scan, not a tool output.

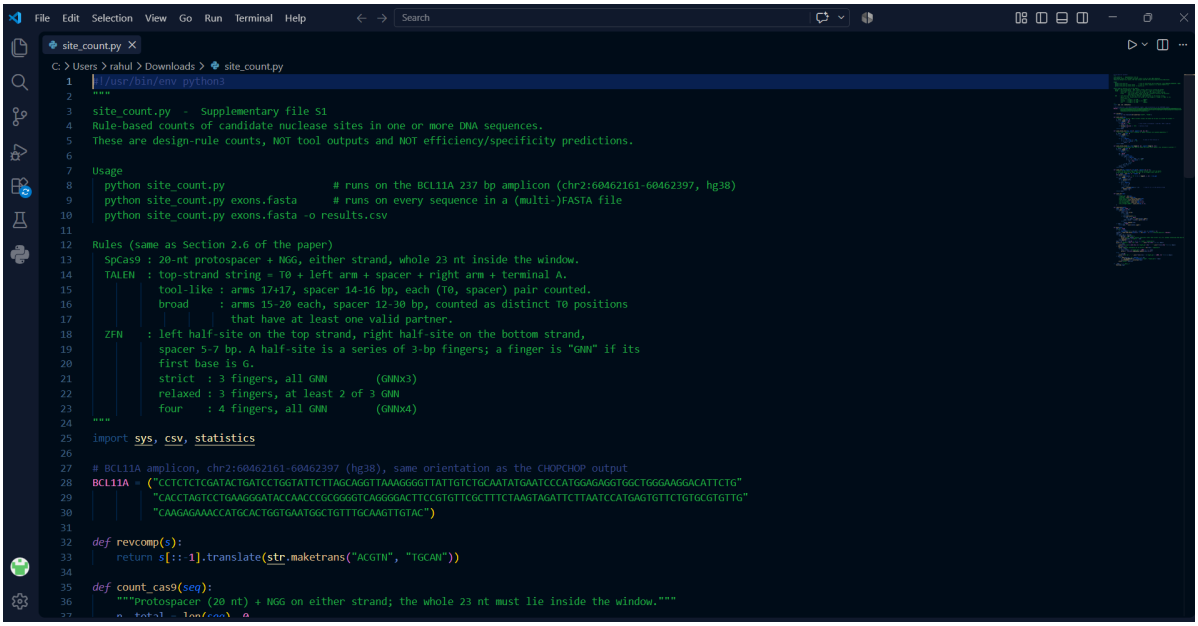


Figure 21. Script S1 (site_count.py) open in Visual Studio Code on the author's laptop (C:/Users/rahul/Downloads/site_count.py). The header documents the usage and the rules of Section 2.6: SpCas9 NGG, TALEN tool-like and broad rules, and ZFN strict, relaxed and four-finger rules.

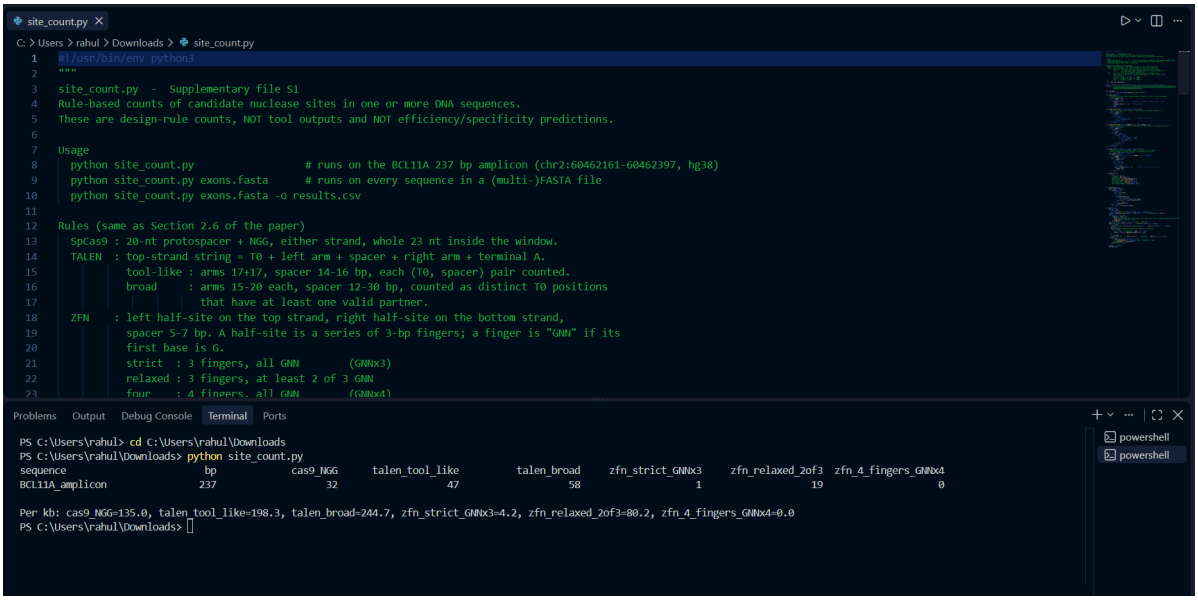
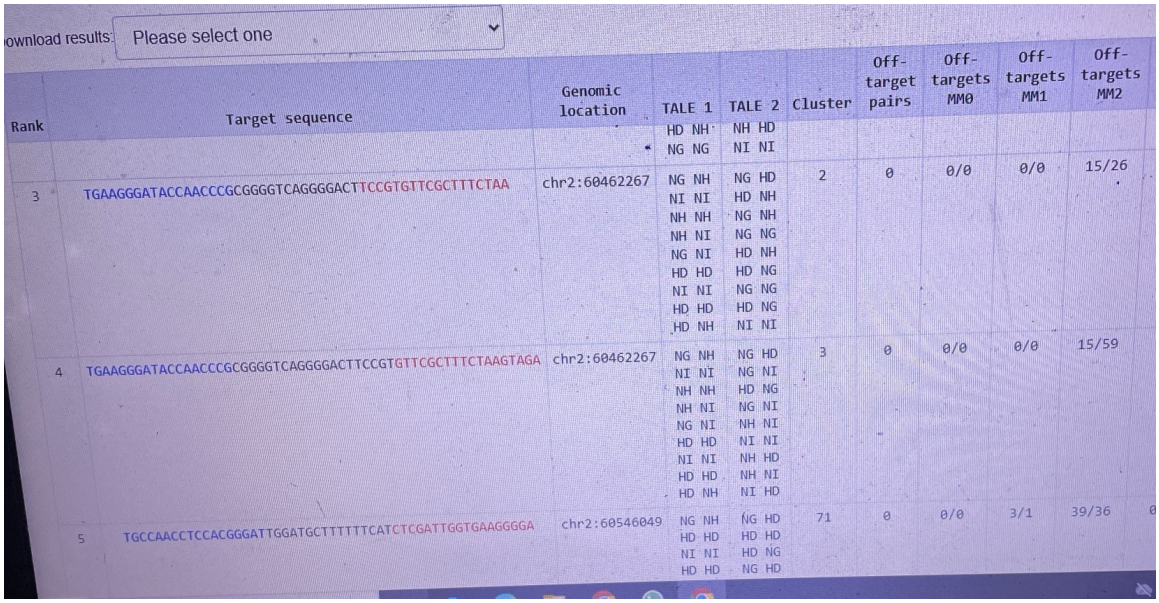


Figure 22. Terminal output of python site_count.py on the BCL11A amplicon (237 bp): SpCas9 32, TALEN tool-like 47, TALEN broad 58, ZFN strict 1, ZFN relaxed 19, ZFN four fingers 0, with the per-kb values 135.0, 198.3, 244.7, 4.2, 80.2 and 0.0. The output reproduces every single-window count in Table 11.



Supplementary Figure S2. Photograph of the CHOPCHOP TALEN results table (Rank 3 row, rows 3-5) taken on 24 October 2026, showing the full 18-repeat TALE 1 and TALE 2 arrays. It is a photograph of the screen, not a screenshot, and is included as evidence for Table 9; replace it with a clean screenshot with the table zoomed out.

6.1 Data provenance notes

- An unrelated university 403 page seen while trying to open a ZFN design tool is not shown: it did not document a tool failure, and the tool it belonged to was never identified.
- The first CRISPOR input-page screenshot (hg19 selected, a stray cut-site marker) is omitted; the results header in Figure 8 confirms that the analysis used hg38.
- The TALEN 'Design plasmid sequences' link did not work and is not cited.

- Primer coordinates and lengths are quoted as reported by CHOPCHOP (for example 60462375-60462397 for a 22-nt primer); the 237-nt sequence string versus the 236-bp product is a counting difference between the systems.
- Screenshots of failed commands during script setup (a script pasted into the terminal, a VS Code extension using a Linux interpreter path, a PowerShell path error) are not shown. The successful run is Figure 22.
- The ZFN pair, all candidate-site counts and the 45-window medians come from the supplementary scripts, not from a design tool.

7. References

Note on verification. References marked with a dagger were added from general knowledge and must be checked against the original journal pages (volume, pages, DOI) before submission. Reference numbers are kept as cited in the text; numbers 46 and 53 are unused and must be closed up when the list is renumbered. Reference [4] (ZFNGenome) was checked against the journal record and is BMC Genomics 2011;12:83.

[1] Kim YG, Cha J, Chandrasegaran S. Hybrid restriction enzymes: zinc finger fusions to Fok I cleavage domain. Proc Natl Acad Sci USA 1996;93:1156-1160.

[2] Wright DA, Thibodeau-Beganny S, Sander JD, et al. Standardized reagents and protocols for engineering zinc finger nucleases by modular assembly. Nat Protoc 2006;1:1637-1652.

[3] Sander JD, Zaback P, Joung JK, Voytas DF, Dobbs D. Zinc Finger Targeter (ZiFiT): an engineered zinc finger/target site design tool. Nucleic Acids Res 2007;35:W599-W605.

[4] Reyon D, Kirkpatrick JR, Sander JD, et al. ZFNGenome: a comprehensive resource for locating zinc finger nuclease target sites in model organisms. BMC Genomics 2011;12:83.

[5] Boch J, Scholze H, Schornack S, et al. Breaking the code of DNA binding specificity of TAL-type III effectors. Science 2009;326:1509-1512.

[6] Moscou MJ, Bogdanove AJ. A simple cipher governs DNA recognition by TAL effectors. Science 2009;326:1501.

[7] Christian M, Cermak T, Doyle EL, et al. Targeting DNA double-strand breaks with TAL effector nucleases. Genetics 2010;186:757-761.

[8] Cermak T, Doyle EL, Christian M, et al. Efficient design and assembly of custom TALEN and other TAL effector-based constructs for DNA targeting. Nucleic Acids Res 2011;39:e82. https://doi.org/10.1093/nar/gkr218

[9] Mussolino C, Morbitzer R, Lütge F, Dannemann N, Lahaye T, Cathomen T. A novel TALE nuclease scaffold enables high genome editing activity in combination with low toxicity. Nucleic Acids Res 2011;39:9283-9293.† https://doi.org/10.1093/nar/gkr597

[10] Sanjana NE, Cong L, Zhou Y, Cunniff MM, Feng G, Zhang F. A transcription activator-like effector (TALE) toolbox for genome engineering. Nat Protoc 2012;7:171-192. https://doi.org/10.1038/nprot.2011.431

[11] Jinek M, Chylinski K, Fonfara I, Hauer M, Doudna JA, Charpentier E. A programmable dual-RNA-guided DNA endonuclease in adaptive bacterial immunity. Science 2012;337:816-821.

[12] Gasiunas G, Barrangou R, Horvath P, Siksnys V. Cas9-crRNA ribonucleoprotein complex mediates specific DNA cleavage for adaptive immunity in bacteria. Proc Natl Acad Sci USA 2012;109:E2579-E2586.

[13] Cong L, Ran FA, Cox D, et al. Multiplex genome engineering using CRISPR/Cas systems. Science 2013;339:819-823.

[14] Mali P, Yang L, Esvelt KM, et al. RNA-guided human genome engineering via Cas9. Science 2013;339:823-826.

[15] Hsu PD, Lander ES, Zhang F. Development and applications of CRISPR-Cas9 for genome engineering. Cell 2014;157:1262-1278.

[16] Labun K, Montague TG, Krause M, Torres Cleuren YN, Tjeldnes H, Valen E. CHOPCHOP v3: expanding the CRISPR web toolbox beyond genome editing. Nucleic Acids Res 2019;47:W171-W174.

[17] Concordet JP, Haeussler M. CRISPOR: intuitive guide selection for CRISPR/Cas9 genome editing experiments and screens. Nucleic Acids Res 2018;46:W242-W245.

[18] Shen MW, Arbab M, Hsu JY, et al. Predictable and precise template-free CRISPR editing of pathogenic variants. Nature 2018;563:646-651.

[19] Doench JG, Fusi N, Sullender M, et al. Optimized sgRNA design to maximize activity and minimize off-target effects of CRISPR-Cas9. Nat Biotechnol 2016;34:184-191.

[20] Hsu PD, Scott DA, Weinstein JA, et al. DNA targeting specificity of RNA-guided Cas9 nucleases. Nat Biotechnol 2013;31:827-832.

[21] Bauer DE, Kamran SC, Lessard S, et al. An erythroid enhancer of BCL11A subject to genetic variation determines fetal hemoglobin level. Science 2013;342:253-257.

[22] Frangoul H, Altshuler D, Cappellini MD, et al. CRISPR-Cas9 gene editing for sickle cell disease and beta-thalassemia. N Engl J Med 2021;384:252-260.

[23] Sankaran VG, Menne TF, Xu J, et al. Human fetal hemoglobin expression is regulated by the developmental stage-specific repressor BCL11A. Science 2008;322:1839-1842.

[24] Canver MC, Smith EC, Sher F, et al. BCL11A enhancer dissection by Cas9-mediated in situ saturating mutagenesis. Nature 2015;527:192-197.

[25] Noël L, Denancé N, Szurek B. Predicting promoters targeted by TAL effectors in plant genomes: from dream to reality. Front Plant Sci 2013;4:333. https://doi.org/10.3389/fpls.2013.00333

[26] Perez-Quintero AL, Szurek B. A decade decoded: spies and hackers in the history of TAL effectors research. Annu Rev Phytopathol 2019;57:459-481.

[27] Chen W, McKenna A, Schreiber J, et al. Massively parallel profiling and predictive modeling of the outcomes of CRISPR/Cas9-mediated double-strand break repair. Nucleic Acids Res 2019;47:7989-8003.

[28] Kleinstiver BP, Pattanayak V, Prew MS, et al. High-fidelity CRISPR-Cas9 nucleases with no detectable genome-wide off-target effects. Nature 2016;529:490-495.

[29] Maeder ML, Thibodeau-Beganny S, Osiak A, et al. Rapid 'open-source' engineering of customized zinc-finger nucleases for highly efficient gene modification. Mol Cell 2008;31:294-301.

[30] Ramirez CL, Foley JE, Wright DA, et al. Unexpected failure rates for modular assembly of engineered zinc fingers. Nat Methods 2008;5:374-375.

[31] Tsai SQ, Zheng Z, Nguyen NT, et al. GUIDE-seq enables genome-wide profiling of off-target cleavage by CRISPR-Cas nucleases. Nat Biotechnol 2015;33:187-197.

[32] Urnov FD, Rebar EJ, Holmes MC, Zhang HS, Gregory PD. Genome editing with engineered zinc finger nucleases. Nat Rev Genet 2010;11:636-646.

[33] Bogdanove AJ, Voytas DF. TAL effectors: customizable proteins for DNA targeting. Science 2011;333:1843-1846.

[34] Doudna JA, Charpentier E. The new frontier of genome engineering with CRISPR-Cas9. Science 2014;346:1258096.

[35] Qasim W, Zhan H, Samarasinghe S, et al. Molecular remission of infant B-ALL after infusion of universal TALEN gene-edited CAR T cells. Sci Transl Med 2017;9:eaaj2013.

[36] Chen KL, Gao C. TALENs: customizable molecular DNA scissors for genome engineering of plants. J Genet Genomics 2013;40:271-279. https://doi.org/10.1016/j.jgg.2013.03.009

[37] Khan Z, Khan SH, Ahmad A, et al. Use of TALEs and TALEN technology for genetic improvement of plants. Plant Mol Biol Rep 2017;35:1-19.

[38] Razzaq A, Saleem F, Kanwal M, et al. Modern trends in plant genome editing: an inclusive review of the CRISPR/Cas9 toolbox. Int J Mol Sci 2019;20:4045.

[39] Hacein-Bey-Abina S, Von Kalle C, Schmidt M, et al. LMO2-associated clonal T cell proliferation in two patients after gene therapy for SCID-X1. Science 2003;302:415-419.

[40] Haapaniemi E, Botla S, Persson J, Schmierer B, Taipale J. CRISPR-Cas9 genome editing induces a p53-mediated DNA damage response. Nat Med 2018;24:927-930.†

[41] Kosicki M, Tomberg K, Bradley A. Repair of double-strand breaks induced by CRISPR-Cas9 leads to large deletions and complex rearrangements. Nat Biotechnol 2018;36:765-771.†

[42] Ran FA, Hsu PD, Lin CY, et al. Double nicking by RNA-guided CRISPR Cas9 for enhanced genome editing specificity. Cell 2013;154:1380-1389.†

[43] Miller JC, Holmes MC, Wang J, et al. An improved zinc-finger nuclease architecture for highly specific genome editing. Nat Biotechnol 2007;25:778-785.†

[44] Wang Y, Cheng X, Shan Q, et al. Simultaneous editing of three homoeoalleles in hexaploid bread wheat confers heritable resistance to powdery mildew. Nat Biotechnol 2014;32:947-951. https://doi.org/10.1038/nbt.2969

[45] Kazama T, Okuno M, Watari Y, et al. Curing cytoplasmic male sterility via TALEN-mediated mitochondrial genome editing. Nat Plants 2019;5:722-730. https://doi.org/10.1038/s41477-019-0459-z

[47] Zhang D, Pries V, Boch J. Targeted C-G-to-T-A base editing with TALE-cytosine deaminases in plants. BMC Biol 2024;22. https://doi.org/10.1186/s12915-024-01895-0

[48] Charlesworth CT, Deshpande PS, Dever DP, et al. Identification of preexisting adaptive immunity to Cas9 proteins in humans. Nat Med 2019;25:249-254.†

[49] Ihry RJ, Worringer KA, Salick MR, et al. p53 inhibits CRISPR-Cas9 engineering in human pluripotent stem cells. Nat Med 2018;24:939-946.†

[50] Tsai SQ, Nguyen NT, Malagon-Lopez J, et al. CIRCLE-seq: a highly sensitive in vitro screen for genome-wide CRISPR-Cas9 nuclease off-targets. Nat Methods 2017;14:607-614.†

[51] Kim D, Bae S, Park J, et al. Digenome-seq: genome-wide profiling of CRISPR-Cas9 off-target effects in human cells. Nat Methods 2015;12:237-243.†

[52] Chen JS, Dagdas YS, Kleinstiver BP, et al. Enhanced proofreading governs CRISPR-Cas9 targeting accuracy. Nature 2017;550:407-410.†

[54] Montague TG, Cruz JM, Gagnon JA, Church GM, Valen E. CHOPCHOP: a CRISPR/Cas9 and TALEN web tool for genome editing. Nucleic Acids Res 2014;42:W401-W407.†

[55] Doyle EL, Booher NJ, Standage DS, et al. TAL Effector-Nucleotide Targeter (TALE-NT) 2.0: tools for TAL effector design and target prediction. Nucleic Acids Res 2012;40:W117-W122.†

Supplementary material: script listings

S1: site_count.py (rule-based counts; reproduces Table 11 with no arguments and the 45-window medians with a FASTA file). S2: fetch_windows.py (downloads the central 237 bp of each listed region from the UCSC sequence service). S3: make_exons.py (BED12 transcripts to unique coding exons). The files accompany this manuscript.

S1: site_count.py

```
#!/usr/bin/env python3
"""
site_count.py  -  Supplementary file S1
Rule-based counts of candidate nuclease sites in one or more DNA sequences.
These are design-rule counts, NOT tool outputs and NOT efficiency/specificity predictions.

Usage
  python site_count.py                 # runs on the BCL11A 237 bp amplicon (chr2:60462161-60462397, hg38)
  python site_count.py exons.fasta     # runs on every sequence in a (multi-)FASTA file
  python site_count.py exons.fasta -o results.csv

Rules (same as Section 2.6 of the paper)
  SpCas9 : 20-nt protospacer + NGG, either strand, whole 23 nt inside the window.
  TALEN  : top-strand string = T0 + left arm + spacer + right arm + terminal A.
           tool-like : arms 17+17, spacer 14-16 bp, each (T0, spacer) pair counted.
           broad     : arms 15-20 each, spacer 12-30 bp, counted as distinct T0 positions
                       that have at least one valid partner.
  ZFN    : left half-site on the top strand, right half-site on the bottom strand,
           spacer 5-7 bp. A half-site is a series of 3-bp fingers; a finger is "GNN" if its
           first base is G.
           strict  : 3 fingers, all GNN        (GNNx3)
           relaxed : 3 fingers, at least 2 of 3 GNN
           four    : 4 fingers, all GNN        (GNNx4)
"""
import sys, csv, statistics

# BCL11A amplicon, chr2:60462161-60462397 (hg38), same orientation as the CHOPCHOP output
BCL11A = ("CCTCTCTCGATACTGATCCTGGTATTCTTAGCAGGTTAAAGGGGTTATTGTCTGCAATATGAATCCCATGGAGAGGTGGCTGGGAAGGACATTCTG"
          "CACCTAGTCCTGAAGGGATACCAACCCGCGGGGTCAGGGGACTTCCGTGTTCGCTTTCTAAGTAGATTCTTAATCCATGAGTGTTCTGTGCGTGTTG"
          "CAAGAGAAACCATGCACTGGTGAATGGCTGTTTGCAAGTTGTAC")

def revcomp(s):
    return s[::-1].translate(str.maketrans("ACGTN", "TGCAN"))

def count_cas9(seq):
    """Protospacer (20 nt) + NGG on either strand; the whole 23 nt must lie inside the window."""
    n, total = len(seq), 0
    for i in range(n - 22):
        w = seq[i:i + 23]
        if w[21:23] == "GG":            # top strand: protospacer = w[0:20], PAM = w[20:23]
            total += 1
        if revcomp(w)[21:23] == "GG":   # bottom strand
            total += 1
    return total

def count_talen_tool(seq, arm=17, spacers=(14, 15, 16)):
    """T0 + arm + spacer + arm + terminal A. Spacer variants are counted separately."""
    n, total = len(seq), 0
    for i in range(n):
        if seq[i] != "T":
            continue
        for s in spacers:
            j = i + 1 + arm + s + arm          # index of the terminal A
            if j < n and seq[j] == "A":
                total += 1
    return total

def count_talen_broad(seq, arms=range(15, 21), spacers=range(12, 31)):
    """Distinct T0 positions that have at least one valid (arm, spacer, arm, terminal A) partner."""
    n, total = len(seq), 0
    for i in range(n):
        if seq[i] != "T":
            continue
        ok = False
        for left in arms:
            for right in arms:
                for s in spacers:
                    j = i + 1 + left + s + right
                    if j < n and seq[j] == "A":
                        ok = True
                        break
                if ok: break
```

```
            if ok: break
        total += ok
    return total

def count_zfn(seq, fingers=3, min_gnn=3, spacers=(5, 6, 7)):
    """Pairs of half-sites (top strand + bottom strand) with a 5-7 bp spacer.
    A half-site passes if at least `min_gnn` of its fingers start with G."""
    n, L, total = len(seq), 3 * fingers, 0
    def passes(site):
        return sum(site[k] == "G" for k in range(0, L, 3)) >= min_gnn
    for i in range(n - L + 1):
        left = seq[i:i + L]
        if not passes(left):
            continue
        for sp in spacers:
            j = i + L + sp
            if j + L > n:
                continue
            right = revcomp(seq[j:j + L])       # read 5'->3' on the bottom strand
            if passes(right):
                total += 1
    return total

def count_all(seq):
    return {
        "length_bp": len(seq),
        "cas9_NGG": count_cas9(seq),
        "talen_tool_like": count_talen_tool(seq),
        "talen_broad": count_talen_broad(seq),
        "zfn_strict_GNNx3": count_zfn(seq, 3, 3),
        "zfn_relaxed_2of3": count_zfn(seq, 3, 2),
        "zfn_4_fingers_GNNx4": count_zfn(seq, 4, 4),
    }

def read_fasta(path):
    name, chunks = None, []
    with open(path) as fh:
        for line in fh:
            line = line.strip()
            if not line:
                continue
            if line.startswith(">"):
                if name is not None:
                    yield name, "".join(chunks).upper()
                name, chunks = line[1:].split()[0], []
            else:
                chunks.append(line)
    if name is not None:
        yield name, "".join(chunks).upper()

def main(argv):
    out_csv = None
    if "-o" in argv:
        k = argv.index("-o"); out_csv = argv[k + 1]; del argv[k:k + 2]
    records = list(read_fasta(argv[1])) if len(argv) > 1 else [("BCL11A_amplicon", BCL11A)]
    rows = []
    for name, seq in records:
        if set(seq) - set("ACGT"):
            print(f"note: {name} contains characters other than A/C/G/T (e.g. N); windows containing them may be misco
      unted")
        r = count_all(seq); r["name"] = name
        rows.append(r)
    keys = [k for k in rows[0] if k not in ("name", "length_bp")]
    print(f"{'sequence':28s} {'bp':>5s} " + " ".join(f"{k:>20s}" for k in keys))
    for r in rows:
        print(f"{r['name'][:28]:28s} {r['length_bp']:5d} " + " ".join(f"{r[k]:20d}" for k in keys))
    if len(rows) > 1:
        print("\nMedian candidates per kb across", len(rows), "sequences")
        for k in keys:
            vals = [r[k] / r["length_bp"] * 1000 for r in rows]
            print(f"  {k:22s} {statistics.median(vals):8.1f}")
    else:
        r = rows[0]
        print("\nPer kb:", ", ".join(f"{k}={r[k] / r['length_bp'] * 1000:.1f}" for k in keys))
    if out_csv:
        with open(out_csv, "w", newline="") as fh:
            w = csv.DictWriter(fh, fieldnames=["name", "length_bp"] + keys)
            w.writeheader(); w.writerows(rows)
        print("wrote", out_csv)

if __name__ == "__main__":
    main(sys.argv)
```

S2: fetch_windows.py

```
#!/usr/bin/env python3
"""
fetch_windows.py - download DNA windows from the UCSC sequence service and write a FASTA file.

Input: a text file with one region per line (tab or space separated):
        chrom  start  end  name        e.g.  chr2  60462160  60462397  BCL11A_amp
       start/end are 0-based half-open, exactly as in a UCSC BED file.
Each region longer than WINDOW bp is cut to its central WINDOW bp; shorter regions are skipped,
so every sequence has the same length (default 237 bp, the length of the BCL11A amplicon).

Usage:  python fetch_windows.py regions.txt exons.fasta [window_bp]
Needs an internet connection. Not run in the sandbox where this was written: test it on one line first.
"""
import sys, json, time, urllib.request

API = "https://api.genome.ucsc.edu/getData/sequence?genome=hg38;chrom={c};start={s};end={e}"

def fetch(chrom, start, end):
    with urllib.request.urlopen(API.format(c=chrom, s=start, e=end), timeout=30) as r:
        return json.load(r)["dna"].upper()

def main():
    regions, out = sys.argv[1], sys.argv[2]
    window = int(sys.argv[3]) if len(sys.argv) > 3 else 237
    kept = skipped = 0
    with open(regions) as fh, open(out, "w") as fo:
        for line in fh:
            parts = line.split()
            if len(parts) < 3 or line.startswith(("#", "track", "browser")):
                continue
            chrom, start, end = parts[0], int(parts[1]), int(parts[2])
            name = parts[3] if len(parts) > 3 else f"{chrom}_{start}_{end}"
            if end - start < window:
                skipped += 1; continue
            mid = (start + end) // 2
            s = mid - window // 2
            e = s + window
            seq = fetch(chrom, s, e)
            fo.write(f">{name}_{chrom}:{s}-{e}\n{seq}\n")
            kept += 1
            time.sleep(0.3)          # be polite to the server
    print(f"wrote {kept} windows to {out}; skipped {skipped} regions shorter than {window} bp")

if __name__ == "__main__":
    main()
```

S3: make_exons.py

```
#!/usr/bin/env python3
"""
make_exons.py - turn a UCSC RefSeq BED12 (one record per transcript) into a
BED4 list of unique protein-coding exons (CDS part only).

Usage: python make_exons.py exon.bed exons.bed

Filters (state these in the paper):
  - curated protein-coding transcripts only (accession starts with NM_)
  - standard chromosomes only (chr1-22, X, Y); no alt/random/fix contigs
  - only the coding part of each exon (clipped to thickStart..thickEnd)
  - identical exon coordinates from different transcripts counted once
"""
import sys, re

STD = {f"chr{i}" for i in range(1, 23)} | {"chrX", "chrY"}

def main(inp, out):
    seen, n_tx, n_nm = set(), 0, 0
    rows = []
    for line in open(inp):
        if line.startswith(("track", "browser", "#")) or not line.strip():
            continue
        f = line.rstrip("\n").split("\t")
        if len(f) < 12:
            sys.exit("This is not a BED12 file (need 12 columns).")
        n_tx += 1
        chrom, name = f[0], f[3]
        if chrom not in STD or not name.startswith("NM_"):
            continue
        n_nm += 1
        start = int(f[1]); tS = int(f[6]); tE = int(f[7])
        sizes = [int(x) for x in f[10].strip(",").split(",")]
        offs = [int(x) for x in f[11].strip(",").split(",")]
        for s, o in zip(sizes, offs):
            a, b = start + o, start + o + s
            a, b = max(a, tS), min(b, tE)      # coding part only
```

```
            if b <= a:
                continue
            key = (chrom, a, b)
            if key in seen:
                continue
            seen.add(key)
            rows.append((chrom, a, b, name))
    with open(out, "w") as w:
        for c, a, b, n in rows:
            w.write(f"{c}\t{a}\t{b}\t{n}\n")
    ge = sum(1 for r in rows if r[2] - r[1] >= 237)
    print(f"transcripts read: {n_tx}; NM_ on standard chromosomes: {n_nm}")
    print(f"unique coding exons written: {len(rows)}; of these >=237 bp: {ge}")

if __name__ == "__main__":
    if len(sys.argv) != 3:
        sys.exit(__doc__)
    main(sys.argv[1], sys.argv[2])
```

Run commands used

```
python site_count.py                                    # BCL11A amplicon: 32, 47, 58, 1, 19, 0
python fetch_windows.py test.txt test.fasta             # one-line test region (chr2 60462160 60462397)
python make_exons.py exon.bed exons.bed                 # 216,361 unique coding exons; 23,754 >= 237 bp
python -c "import random;L=[l for l in open('exons.bed')];random.seed(1);open('regions.txt','w').writelines(random.sample(L,500))"
python fetch_windows.py regions.txt exons.fasta         # 45 windows kept
python site_count.py exons.fasta -o results.csv         # medians per kb across windows
```